Morgan Stanley | RESEARCH

IDEA

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Greater China Semiconductors | Asia Pacific

The Foundry Floorplan: All boats riding on the same tide – mature node up-cycle begins

AI power ICs sit at the core of future AI infrastructure and are growing as fast as, if not faster than, AI GPUs. We therefore expect a mature node capacity shortage in 2H27. We raise price targets for mature node foundries and upgrade UMC to OW.

Mature node up-cycle coming: With AI infrastructure build-outs likely to last two more years, some smartphone/PC customers now worry about mature node shortages. SMIC and UMC flagged similar trends in their recent earnings calls. So, we don't see inventory corrections or order cuts at mature foundries any time soon.

AI power IC foundry opportunity: Global leaders, including Infineon and MPS, are partly adopting a foundry outsourcing model. For example, MPS uses both Hua Hong and Vanguard; Infineon uses UMC. We estimate AI server power IC TAM at US $15bn in 2026, with a 50% CAGR over the next two years, rising with global AI GPU demand. AI power ICs should more than offset weak consumer and smartphone semi demand. One of Hua Hong's major US PMIC customers is tripling wafer demand on its 55nm BCD platform.

Key changes within this report:

- **Upgrade UMC to OW:** We cite attractive valuation and earnings upside from stronger mature node demand. P/E is 20x for 2027e, vs. Vanguard at 27x.
- **Keep SMIC OW and raise price target:** We like its strong BCD process at the new 8-inch fab and aggressive advanced node expansion.
- **Raise price targets but stay EW on Hua Hong and Vanguard:** Valuation looks fair after the stocks' rally.

Segments on which we're turning cautious:

- **Consumer/automotive PMIC:** We downgrade Silergy to UW given margin miss from foundry cost hikes and weak consumer demand. Its server and optical PMIC exposure remains relatively small. (See 1Q26 earnings review.)
- **Downgrade GWC to EW:** Expectations for raw wafer price hikes look a bit high for 2H26. Revenue upside should be largely priced in.

What about TSMC and our thoughts about Intel foundry competition? Recent newsflow suggests that Intel foundry is engaging with Tesla and Apple. Tesla has been working with Intel on a TeraFab foundry partnership and Intel foundry also produced Apple's iPhone modem in 2018, so trials at Intel foundry, such as AI6 chips, could be possible. We believe it will take time for yield to ramp, which has limited impact on TSMC leading-edge in the near term.

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GREATER CHINA TECHNOLOGY SEMICONDUCTORS

Asia Pacific
Industry View **Attractive**

WHAT'S CHANGED

	From	To
UMC (2303.TW)		
Rating	Equal-weight	**Overweight**
Price Target	NT$68.00	**NT$138.00**
Vanguard International Semiconductor (5347.TWO)	From	To
Price Target	NT$148.00	**NT$180.00**
GlobalWafers Co Ltd (6488.TWO)	From	To
Price Target	NT$588.00	**NT$750.00**
Silergy Corp. (6415.TW)	From	To
Rating	Equal-weight	**Underweight**
SMIC (0981.HK)	From	To
Price Target	HK$70.00	**HK$85.00**
Hua Hong Semiconductor Ltd (1347.HK)	From	To

Price Target	HK$88.00	**HK$118.00**

We see expanding foundry utilization and revenue growth for both leading-edge and mature nodes

TSMC comments on mature node strategy – support for mature node customers will not be compromised in the near term

TSMC recently commented on its mature node strategy, indicating a more disciplined approach toward legacy capacity management as the company continues to prioritize investment in leading-edge technologies and advanced packaging expansion driven by AI demand. Management noted that TSMC will take a few years to gradually wind down Fab 2 (6-inch) and Fab 5 (GaN) in order to optimize land utilization and engineering resources. That said, the company emphasized that capacity support for mature node customers will not be compromised during the transition.

We believe this reflects TSMC's broader strategy of reallocating resources toward higher-growth and higher-return businesses, while still maintaining long-term customer support for legacy applications. At the same time, tighter mature node capacity management at leading foundries could create incremental overflow opportunities for specialty and second-tier foundry players in the medium to long term.

Mature node foundries benefit from strong AI/memory-related demand and potential overflow from TSMC and Samsung in the medium to long term

Both TSMC and Samsung have been gradually consolidating and rationalizing their mature node capacity as they prioritize allocation of capital and resources toward leading-edge technologies and advanced packaging for AI-related demand. Thus, they have become more selective about availability of mature node capacity, particularly for specialty and lower-volume applications.

We believe this could create overflow demand opportunities for second-tier foundries and specialty players in the medium to long term, as customers seek alternative manufacturing partners to secure stable supply and cost-efficient production. This trend could benefit foundries with established mature node capabilities, especially in areas such as power management ICs, display drivers, connectivity chips, and industrial semiconductors.

PSMC is one of the key beneficiaries within our coverage amid recent historical memory price hikes. For the traditional memory foundry business, PSMC continues to engage in monthly price negotiations with customers, and we expect memory foundry price hike to continue. See more details from our PSMC report.

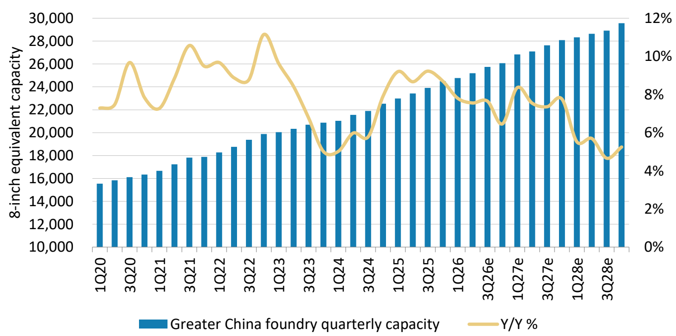
UMC is also hiking wafer prices in response to demand recovery. We now assume a 5-10% average wafer price hike in 2H26, and more to come in 2027. We also expect the capacity at UMC to remain relatively tight from 2H26 to 2028, given the strong AI peripheral demand.

TSMC is outsourcing interposer production to Vanguard given insufficient capacity

During Vanguard's latest earnings call, management stated that TSMC is outsourcing a portion of its interposer production to Vanguard because demand for advanced packaging continues to outpace available internal capacity. The move reflects the industry-wide bottleneck in CoWoS and related advanced packaging technologies, driven primarily by accelerating AI and high-performance computing demand.

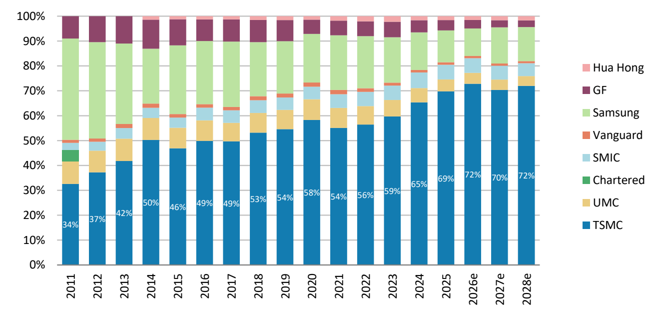
By leveraging Vanguard's mature manufacturing capabilities in Singapore, TSMC aims to alleviate supply constraints in 2027, improve production flexibility, and support key customers requiring advanced chip integration solutions. This partnership leads to higher commitment to and utilization of Vanguard's fab in Singapore. Our checks also suggest that Vanguard's 8-inch wafer price hike could range from the low single digits to the low teens (%) in 1H26. It is even raising prices for LPDDIC given tight capacity, which appears to adopt a more mature process.

Exhibit 1: Greater China foundry capacity could grow in high single digits Y/Y in 2026



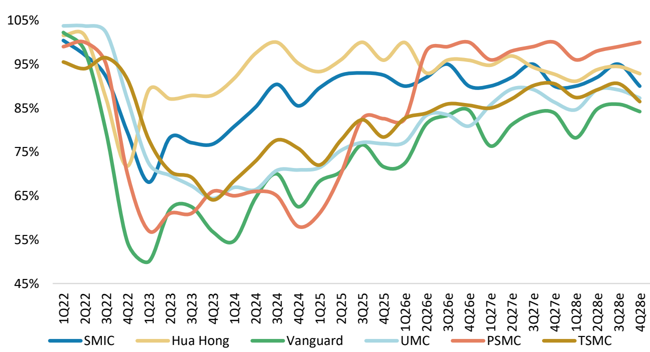
Source: Morgan Stanley Research estimates

Exhibit 2: TSMC accounts for the majority of the foundry industry



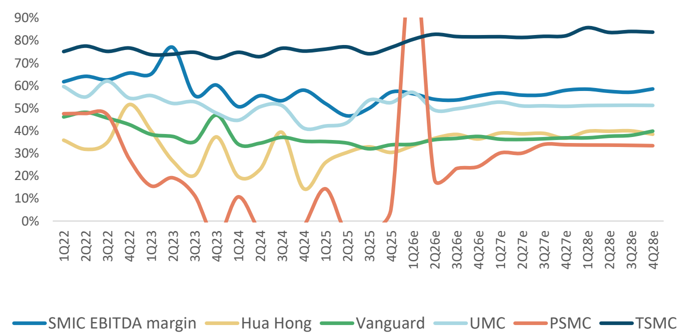
Source: Morgan Stanley Research estimates

Exhibit 3: Overall utilization rate should recover gradually for Greater China foundry



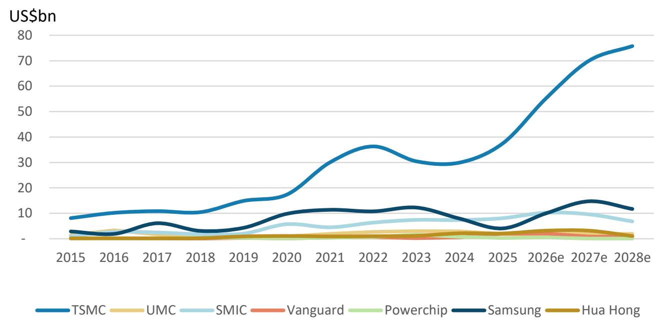
Source: Corporate Data, Morgan Stanley Research estimates

Exhibit 4: TSMC's EBITDA margin remains stable at 75-80%



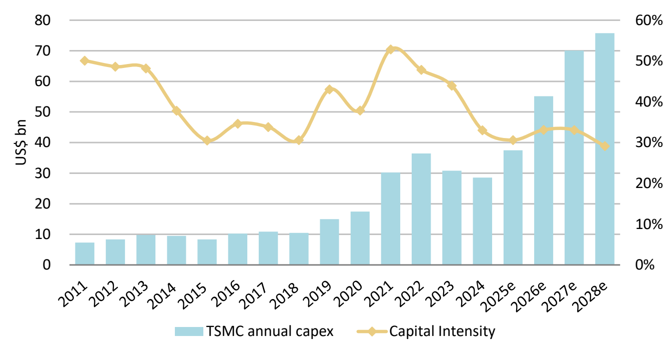
Source: Corporate Data, Morgan Stanley Research estimates

Exhibit 5: TSMC leads capex for the whole industry



Source: Company Data, Morgan Stanley Research estimates

Exhibit 6: TSMC's capex intensity should drop significantly from robust revenue growth



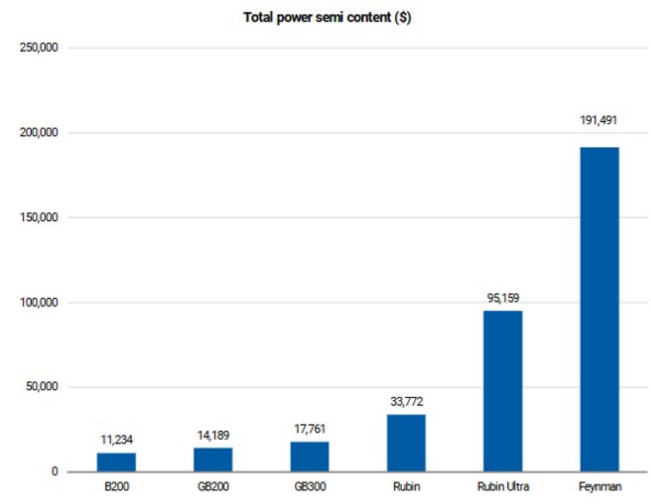
Source: Company Data, Morgan Stanley Research estimates

Structural improvement for power IC demand – data center acceleration and the 800V transition

Beyond AI accelerators themselves, AI-driven demand is also fueling strong growth across the broader semiconductor ecosystem, including CPUs, power management ICs, networking chips, and advanced memory solutions. As hyperscalers and enterprises continue to expand AI infrastructure deployments, demand for high-performance server CPUs, high-bandwidth networking silicon, and increasingly sophisticated power delivery architectures has accelerated meaningfully. In particular, rising GPU and AI server power consumption is driving significant content growth for power IC suppliers.

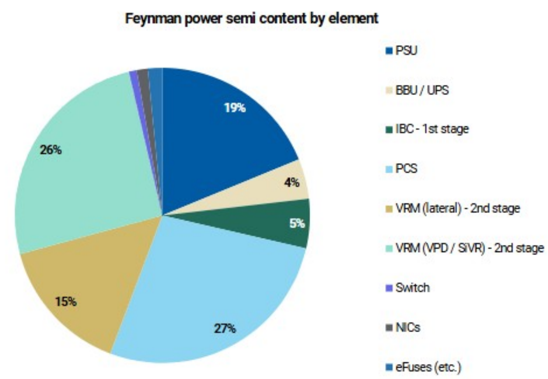
Our Global Tech analyst, Shawn Kim, in his report, shared the same view: that AI/data centers are turning analog semis from a pure auto/industrial cycle into a power and connectivity content story. He believes the most important structural shift for analog is the next 12-24 months of AI rack power evolution, especially the move toward 800V architectures. Nvidia's Kyber rack is the lead example. Our European semis team has increased its assessment of server rack model granularity to incorporate a PCS (Power Capacitor System) inside the new power sidecar. The sidecar absorbs parts of the traditional PSU (Power Supply Unit), rack-level BBU (Battery Back-up Unit), and bulk capacitance previously inside servers. Morgan Stanley estimates more than US$20k of power-semiconductor content for the PCS per Rubin Ultra rack, with total power-semiconductor BoM rising to US$159/kW for Rubin Ultra and US$191/kW for Feynman.

Exhibit 7: Morgan Stanley estimates power semi content in Feynman to reach US$191/kW.



Source:
Morgan Stanley Research estimates

Exhibit 8: Morgan Stanley sees the PCS, VRM (VPD/SVR) and VRM (lateral) as the highest US$-content power semi components in the Feynman rack



Source:
Morgan Stanley Research estimates

Read-across from mature node companies' 1Q26 earnings

SMIC: AI spillover story

AI demand is broadening beyond computing silicon: SMIC's call reinforces our view that the AI cycle is increasingly spilling over from leading-edge computing into mature node companion silicon (AI-driven PMIC demand supporting specialty node utilization). The company framed its AI-related opportunity around power management, edge AI logic, connectivity, etc, rather than the AI accelerator itself. This distinction matters: the incremental demand described by management is not a direct GPU story, but a mature node and specialty process story, where AI servers, edge AI, robotics, and AI-enabled consumer devices require a growing volume of analog, mixed-signal, power, and control chips.

Capacity reallocation is creating structural tightness in non-leading-edge products: Management's comments suggest that AI is acting as a global capacity "siphon": foundries and IDMs are prioritising AI computing, HBM, and higher-margin memory, which in turn reduces available capacity for standard logic, networking, and niche memory products. This is driving order reshoring and capacity seeking into mainland China.

Specialty memory is becoming a second-order beneficiary of the AI memory cycle: The memory read-through is not that SMIC benefits directly from DRAM or HBM pricing, but that tightness in mainstream memory is cascading into SLC NAND, NOR Flash, and other specialty memory products. SMIC's advantage is process flexibility: specialty memory shares a high degree of equipment and process overlap with existing logic capacity, enabling gradual conversion with limited incremental dedicated equipment.

8-inch BCD remains the bottleneck: SMIC's 8-inch revenue growth was driven mainly by pricing and mix, not new capacity, as the platform has been effectively full. The areas of stronger demand include AI-related PMIC and power delivery, automotive-grade analog BCD, embedded memory, and selected sensing/display technologies such as ToF and Micro OLED.

Pricing tailwinds should help offset depreciation pressure: The pricing story is selective. Management is negotiating price increases in tight product categories, with the impact expected to become more visible through 2Q and continue into 2H. This should support gross margin alongside richer product mix and higher shipments. However, depreciation remains a material offset, with full-year depreciation expected to rise sharply.

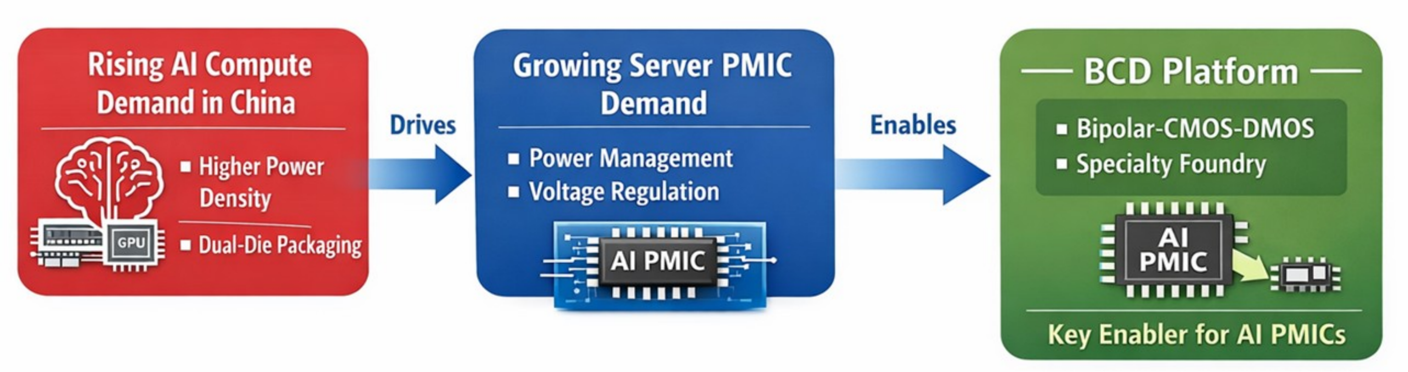
Hua Hong: PMIC tightness is the core AI read-through

AI demand is translating most clearly into PMIC and BCD: Hua Hong's AI read-through is concentrated in PMIC, largely built on its BCD platform and specialty technology. Demand is driven by AI servers, automotive electronics, and robotics, while consumer remains soft. PMIC already grew in 1Q but was capacity-constrained.

NOR Flash is benefiting from memory-cycle spillover: Hua Hong's memory exposure is mainly embedded and standalone non-volatile memory, especially NOR Flash, rather than DRAM. DRAM tightness is spilling into NAND and then NOR, lifting NOR demand and pricing. This gives Hua Hong exposure to the memory up-cycle through specialty memory pricing and utilization, rather than direct HBM/DRAM beta.

Power discretes are healthy, but less pricing-sensitive: AI servers and industrial demand support power discretes, but supply is also expanding, especially as SiC gains share. This makes the segment more competitive than PMIC/BCD. Hua Hong's focus is therefore portfolio broadening, including SiC and GaN, rather than relying on near-term scarcity pricing.

Exhibit 9: BCD platform could be the key enabler of AI-related PMICs



Source: Morgan Stanley Research

UMC: Better mature node recovery supports upgrade to Overweight

UMC's 1Q26 earnings call pointed to a better near-term demand recovery (Non-AI demand recovery surprise to the upside): 2Q26 wafer shipments were guided to grow in high single digits QoQ, ahead of our prior 5% QoQ forecast. Management cited recovery across communications, consumer, and automotive/industrial applications.

UMC also highlighted its capabilities in bridge dies and deep trench capacitors: We believe this could create collaboration opportunities with other foundries, particularly Intel, for large reticle-size chip designs. (See our report, Foundry Floorplan on Intel's EMIB-T.) We now assume a 5-10% price increase in 2H26 and expect utilization to remain tight from 2H26 and beyond.

We upgrade UMC to Overweight – mature node recovery appears stronger and broader than expected. The upgrade is supported by four factors:

1. Stronger shipment recovery, with 2Q26 wafer shipments guided ahead of our prior forecast;
2. 2027 pricing upside – we now expect a 5-10% wafer price hike in 2H26 and more to come in 2027;
3. Improving mix, led by 22nm/28nm and specialty processes; and
4. Emerging bridge die and DTC optionality linked to advanced packaging and large reticle-size chip designs.

Vanguard: Better visibility, but maintain Equal-weight

Vanguard's 1Q26 call also points to a solid near-term recovery (High utilization with interposer production in 2027): However, we maintain our Equal-weight view. Management guided for 2Q26 wafer shipments to increase 11-13% QoQ, driven by AI data center demand and smartphone share gains, while ASP is expected to rise 2-4% through wafer price hikes. This implies 2Q26 revenue growth could approach 15% QoQ, suggesting improving utilization and a healthier mature node demand backdrop.

Management also confirmed that Vanguard's upcoming Singapore 12-inch capacity will include interposer production, with technology transferred from TSMC: This is consistent with our prior checks that Vanguard has been working with TSMC on interposer-related production. We view this as a positive longer-term strategic development, as interposer demand should benefit from advanced packaging and AI infrastructure growth.

Order visibility has improved versus three months ago: Management expects Vanguard to outgrow the overall mature node industry's mid-teens (%) YoY growth in 2026, supported by demand from multiple customers. However, after the recent recovery in sentiment, we think risk-reward is more balanced. The company has better growth visibility and potential interposer optionality, but the magnitude and timing of incremental margin upside still need to be proven. We therefore keep Vanguard at Equal-weight.

About Foundry Floorplan

Back in 2011, we published this product – e.g., Asia/Pacific Foundry: The Foundry Floorplan, April 2011 – to offer our analysis of foundry industry dynamics. We think it was a great practice, and have resumed it given that global foundry competition is still active. Foundry supply is also a critical resource for future AI chips, whether in the global market or China's local market.

Below are prior reports from this series.

- The Foundry Floorplan: TSMC preview update; CoWoS vs. EMIB competition (14 Apr 2026)
- The Foundry Floorplan: Raise TSMC's 2027e and 2028e Capex (9 Mar 2026)
- The Foundry Floorplan: Update on TSMC's 2027 capex; interposer outsourcing to Vanguard? (13 Jan 2026)
- Asia/Pacific Foundry: The Foundry Floorplan, April 2011

UMC: Price Target Raised to NT$138 from NT$68; Upgrade to OW

We raise our 2026, 2027, and 2028 earnings estimates 27%, 25%, and 25%, respectively: The increases mainly reflect stronger AI companion demand and our expectation of more overflow consumer demand from other foundry peers gradually consolidating 8-inch capacity. Higher utilization rate assumption also leads our margin estimates to rise.

Exhibit 10: UMC: Estimate revisions

(NT$ mn)	New 2026e	Old 2026e	Diff.%	New 2027e	Old 2027e	Diff.%	New 2028e	Old 2028e	Diff.%
Net sales	271,238	265,214	2%	327,483	294,890	11%	364,804	322,458	13%
Gross profit	82,884	74,616	11%	108,171	87,635	23%	127,311	101,034	26%
Operating profit	55,860	47,629	17%	78,823	59,941	31%	95,748	71,296	34%
Pretax income	65,511	53,834	22%	86,614	69,176	25%	105,539	84,131	25%
Net income	56,406	44,366	27%	71,242	56,947	25%	86,760	69,210	25%
Basic EPS	**4.52**	**3.55**	**27%**	**5.70**	**4.56**	**25%**	**6.95**	**5.54**	**25%**
Diluted EPS	**4.49**	**3.53**	**27%**	**5.67**	**4.54**	**25%**	**6.91**	**5.51**	**25%**
Margins									
Gross margin	30.6%	28.1%		33.0%	29.7%		34.9%	31.3%	
Operating margin	20.6%	18.0%		24.1%	20.3%		26.2%	22.1%	
Pretax margin	24.2%	20.3%		26.4%	23.5%		28.9%	26.1%	
Net margin	20.8%	16.7%		21.8%	19.3%		23.8%	21.5%	

Source: Morgan Stanley Research (e) estimates

Exhibit 11: UMC: Quarterly financials

(NT$ mn)	1Q26	2Q26e	3Q26e	4Q26e	1Q27e	2Q27e	3Q27e	4Q27e	2024	2025	2026e	2027e	2028e
Total Revenues	**61,038**	**68,096**	**71,619**	**70,485**	**78,665**	**82,404**	**83,717**	**82,696**	**232,303**	**237,553**	**271,238**	**327,483**	**364,804**
Sequential Change	*-1.2%*	*11.6%*	*5.2%*	*-1.6%*	*11.6%*	*4.8%*	*1.6%*	*-1.2%*					
Change vs Year Ago	*5.5%*	*15.9%*	*21.1%*	*14.0%*	*28.9%*	*21.0%*	*16.9%*	*17.3%*	*4.4%*	*2.3%*	*14.2%*	*20.7%*	*11.4%*
Cost of Sales	**(43,219)**	**(47,400)**	**(49,397)**	**(48,337)**	**(52,305)**	**(55,420)**	**(56,031)**	**(55,556)**	**(156,649)**	**(168,647)**	**(188,354)**	**(219,312)**	**(237,493)**
Gross Profit	**17,818**	**20,695**	**22,222**	**22,148**	**26,360**	**26,984**	**27,687**	**27,140**	**75,654**	**68,906**	**82,884**	**108,171**	**127,311**
Gross Margin	29.2%	30.4%	31.0%	31.4%	33.5%	32.7%	33.1%	32.8%	32.6%	29.0%	30.6%	33.0%	34.9%
Total Opex	(6,542)	(6,722)	(6,822)	(6,939)	(7,188)	(7,301)	(7,385)	(7,474)	(24,042)	(24,958)	(27,024)	(29,348)	(31,563)
Percent of Revenues	*10.7%*	*9.9%*	*9.5%*	*9.8%*	*9.1%*	*8.9%*	*8.8%*	*9.0%*	*10.3%*	*10.5%*	*10.0%*	*9.0%*	*8.7%*
R&D	(4,575)	(4,625)	(4,635)	(4,645)	(4,650)	(4,700)	(4,750)	(4,800)	(15,616)	(17,725)	(18,481)	(18,900)	(19,860)
Percent of Revenues	*7.5%*	*6.8%*	*6.5%*	*6.6%*	*5.9%*	*5.7%*	*5.7%*	*5.8%*	*6.7%*	*7.5%*	*6.8%*	*5.8%*	*5.4%*
Sales and Marketing	(689)	(769)	(809)	(796)	(888)	(931)	(945)	(934)	(2,701)	(2,433)	(3,063)	(3,698)	(4,013)
Percent of Revenues	*1.1%*	*1.1%*	*1.1%*	*1.1%*	*1.1%*	*1.1%*	*1.1%*	*1.1%*	*1.2%*	*1.0%*	*1.1%*	*1.1%*	*1.1%*
General and Admin	(1,278)	(1,328)	(1,378)	(1,498)	(1,650)	(1,670)	(1,690)	(1,740)	(5,724)	(4,800)	(5,480)	(6,750)	(7,690)
Percent of Revenues	*2.1%*	*1.9%*	*1.9%*	*2.1%*	*2.1%*	*2.0%*	*2.0%*	*2.1%*	*2.5%*	*2.0%*	*2.0%*	*2.1%*	*2.1%*
Operating Income	**11,276**	**13,974**	**15,401**	**15,209**	**19,171**	**19,684**	**20,301**	**19,666**	**51,613**	**43,949**	**55,860**	**78,823**	**95,748**
Percent of Revenues	*18.5%*	*20.5%*	*21.5%*	*21.6%*	*24.4%*	*23.9%*	*24.2%*	*23.8%*	*22.2%*	*18.5%*	*20.6%*	*24.1%*	*26.2%*
Total Non-operating Income(Loss)	**5,367**	**1,088**	**1,598**	**1,598**	**1,798**	**1,898**	**1,998**	**2,098**	**4,607**	**5,699**	**9,651**	**7,791**	**9,791**
Profit Before Taxes	**16,644**	**15,061**	**16,999**	**16,807**	**20,969**	**21,581**	**22,299**	**21,764**	**56,220**	**49,648**	**65,511**	**86,614**	**105,539**
Percent of Revenues	*27.3%*	*22.1%*	*23.7%*	*23.8%*	*26.7%*	*26.2%*	*26.6%*	*26.3%*	*24.2%*	*20.9%*	*24.2%*	*26.4%*	*28.9%*
Change vs Year Ago	*78.1%*	*48.3%*	*16.1%*	*8.4%*	*26.0%*	*43.3%*	*31.2%*	*29.5%*					
Taxes	(527)	(2,711)	(3,060)	(3,025)	(3,774)	(3,885)	(4,014)	(3,918)	(9,113)	(8,113)	(9,323)	(15,591)	(18,997)
Tax Rate	*3.2%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*16.2%*	*16.3%*	*14.2%*	*18.0%*	*18.0%*
Net Income, Cont Ops	16,117	12,350	13,939	13,782	17,195	17,697	18,285	17,847	47,106	41,535	56,188	71,024	86,542
Percent of Revenues	*26.4%*	*18.1%*	*19.5%*	*19.6%*	*21.9%*	*21.5%*	*21.8%*	*21.6%*	*20.3%*	*17.5%*	*20.7%*	*21.7%*	*23.7%*
Minority Interest	54	54	54	54	54	54	54	54	105	182	218	218	218
Reported Income (TW GAAP)	**16,171**	**12,405**	**13,993**	**13,836**	**17,249**	**17,751**	**18,340**	**17,901**	**47,211**	**41,716**	**56,406**	**71,242**	**86,760**
Percent of Revenues	*26.5%*	*18.2%*	*19.5%*	*19.6%*	*21.9%*	*21.5%*	*21.9%*	*21.6%*	*20.3%*	*17.6%*	*20.8%*	*21.8%*	*23.8%*
Change vs Year Ago	*107.9%*	*39.3%*	*-6.6%*	*37.6%*	*6.7%*	*43.1%*	*31.1%*	*29.4%*	*-22.6%*	*-11.6%*	*35.2%*	*26.3%*	*21.8%*
Reported diluted EPS (NT$, TW GAAP)	**1.29**	**0.99**	**1.11**	**1.10**	**1.37**	**1.41**	**1.46**	**1.43**	**3.76**	**3.32**	**4.49**	**5.67**	**6.91**
Reported Basic EPS (NT$, GAAP)	**1.29**	**0.99**	**1.12**	**1.11**	**1.38**	**1.42**	**1.47**	**1.43**	**3.80**	**3.34**	**4.52**	**5.70**	**6.95**

Source: Company data, Morgan Stanley Research (e) estimates

We raise our price target (base case value) to NT$138 and upgrade the stock to OW: We continue to derive our price target from a residual income model because we believe it best represents the value of the company. Besides our estimate increases, we raise our intermediate growth rate from 5.9% to 12.5% given structural changes in the industry's supply/demand dynamics. Other key assumptions are unchanged, including a cost of equity of 9.2% (beta of 1.2, equity risk premium of 6%, and risk-free rate of 2%), and terminal growth rate of 3%.

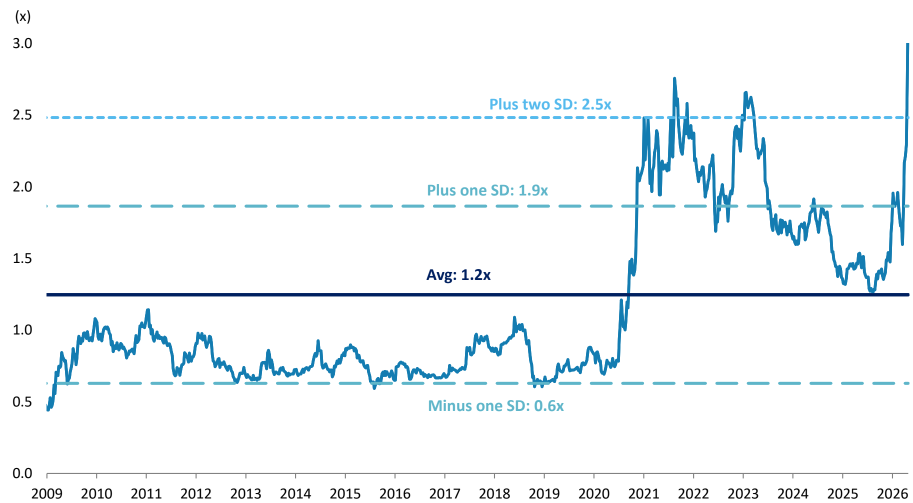
Our bull case value is also raised from NT$97 to NT$200; our bear case value rises from NT$35 to NT$70, implying 5.6x and 2.0x 2027e BVPS, respectively.

Exhibit 12: UMC: Residual income model

(NT$ mn)	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e
Total Equity	**415,188**	**443,457**	**476,001**	**520,899**	**571,410**	**628,234**	**692,162**	**764,080**	**844,989**	**936,010**	**1,038,410**	**1,153,610**
Net Profit	**56,406**	**71,242**	**86,760**	**97,605**	**109,806**	**123,532**	**138,973**	**156,345**	**175,888**	**197,874**	**222,608**	**250,434**
ROAE	**14.2%**	**16.6%**	**18.9%**	**19.6%**	**20.1%**	**20.6%**	**21.1%**	**21.5%**	**21.9%**	**22.2%**	**22.5%**	**22.8%**
Residual Income	18,953	30,699	42,891	49,417	56,805	65,110	74,447	84,944	96,748	110,022	124,950	141,739
Spread	**5.0%**	**7.4%**	**9.7%**	**10.4%**	**10.9%**	**11.4%**	**11.9%**	**12.3%**	**12.7%**	**13.0%**	**13.3%**	**13.6%**
Ending Equity Capital	415,188											
PV of Forecast Period	423,188											
PV of Continuing Value	894,304											
Equity Value	**1,732,679**											
No. of Shares	12,556											
Projected Price (NT$)	**138.0**											

Source: Morgan Stanley Research (e) estimates.

Exhibit 13: UMC's one-year forward P/B band



Source: FactSet, Morgan Stanley Research estimates

Risk Reward – UMC (2303.TW)

Wafer prices on the rise; OW

PRICE TARGET NT$138.00

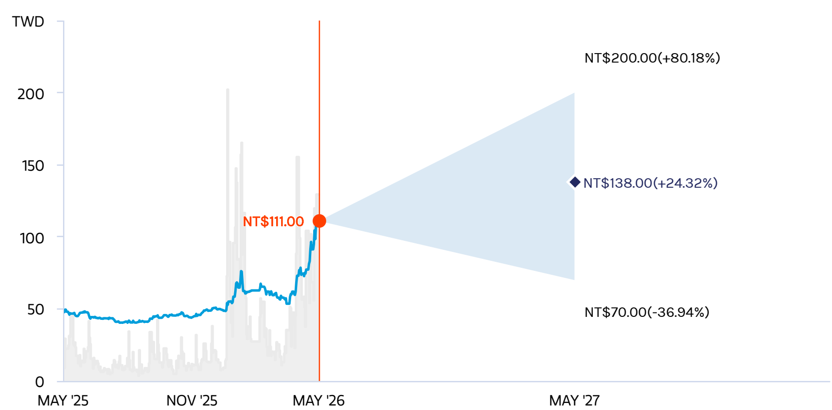
Base case, residual income model. Key assumptions: 9.2% cost of equity, 12.5% medium-term growth, and 3.0% terminal growth.

Consensus Price Target Distribution NT$38.30 — NT$72.41 (Mean) — NT$138.00 (MS PT)

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We have turned OW on UMC. We see structural change in the mature node industry's supply/demand dynamics.
- Its foundry quality is comparable to TSMC's in mature nodes. Along with specialty technologies, this could help offset part of the margin downside.
- Non-AI market recovery remains uncertain in 2026, yet UMC should still benefit from strong AI-related demand spillover from its foundry peers.
- 12nm FinFET development is strategic. It should be one of UMC's long-term strengths, notably when Chinese foundries cannot effectively expand FinFET capacity.
- The stock now trades at 3.3x 2027e BVPS. That is well above its historical average since 2009, but our price target implies 3.9x.

Consensus Rating Distribution

● 23%	Overweight
62%	Equal-weight
15%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era:	*Positive*
Pricing Power:	*Negative*
Secular Growth:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$200.00

5.6x 2027e BVPS

Stabilized semi demand; better-than-expected 8-inch and trailing-edge 12-inch demand: 1) Global semi sales grow >15% Y/Y in 2026, and ex-TSMC foundry industry grows >10% Y/Y; 2) UMC gains share in both 8- and 12-inch wafers thanks to technology differentiation; 3) UMC receives more outsourcing from customers, which helps fill 22/28nm capacity and lifts blended wafer ASP; 4) traction increases for UMC's 12nm/14nm FinFET technologies.

BASE CASE NT$138.00

4.0x 2027e BVPS

Healthy semi growth; improving demand for 8-inch and trailing-edge 12-inch: 1) Global semi sales grow ~15% in 2026 and ex-TSMC foundry industry grows 3-5% Y/Y; 2) 8-inch utilization rate is above 75% with steady 12-inch demand amid semi inventory digestion; 3) UMC hikes wafer prices slightly in response to strong AI-related demand.

BEAR CASE NT$70.00

2.0x 2027e BVPS

Significant semi industry decline; 8-inch and trailing-edge 12-inch face weakening demand: 1) Global semi sales grow only <5% in 2026; 2) display semi and power semi demand deteriorates further; 3) significant wafer pricing erosion given low fab utilization and mature node peers' competition; 4) depreciation ramp-up of more than 30% Y/Y in 2026 without sufficient wafer order inflow to sustain UMC's gross margin.

Risk Reward – UMC (2303.TW)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Wafer shipment (K, 8 inch equivalent) (NT$, mn)	8,741	9,788	10,889	11,207
Wafer ASP (USD) (NT$, mn)	824	861	940	1,018
Utilization rate (%) (%)	75.2	81.2	87.6	87.6

INVESTMENT DRIVERS

- Semiconductor end-demand
- Outsourcing orders from IDM customers
- Ramp-up of new capacity

GLOBAL REVENUE EXPOSURE

- 0-10% Europe ex UK
- 0-10% Japan
- 10-20% Mainland China
- 20-30% North America
- 30-40% APAC, ex Japan, Mainland China and India

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 **MOST** | 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger-than-expected global economic and semiconductor growth
- Looser pricing competition, increasing margins and utilization rates
- Favorable customer mix changes

RISKS TO DOWNSIDE

- Softer-than-expected global economic and semiconductor growth
- Increasing pricing competition, reducing margins and utilization rates
- Unfavorable customer mix changes

OWNERSHIP POSITIONING

Inst. Owners, % Active 72.6%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

	Low	Morgan Stanley Estimates	Mean	High
Sales / Revenue (NT$, mn)	251,852	271,238	269,714	292,702
EBITDA (NT$, mn)	103,573	122,821	121,066	148,490
Net income (NT$, mn)	41,342	56,406	54,668	67,067
EPS (NT$)	3.31	4.49	4.39	5.36

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

UMC: Financial summary

Exhibit 14: UMC: Financial Summary

Income Statement, 2024-28e, Year End Dec

(NT$ mn)	2024	2025	2026e	2027e	2028e
Turnover	**232,303**	**237,553**	**271,238**	**327,483**	**364,804**
YoY Growth	*4.4%*	*2.3%*	*14.2%*	*20.7%*	*11.4%*
Less: COGS	**(156,649)**	**(168,647)**	**(188,354)**	**(219,312)**	**(237,493)**
Variable costs	(111,866)	(114,015)	(125,566)	(151,100)	(169,634)
Depreciation & amort	(44,782)	(54,631)	(62,788)	(68,213)	(67,859)
Gross profit	**75,654**	**68,906**	**82,884**	**108,171**	**127,311**
YoY Growth	*-2.7%*	*-8.9%*	*20.3%*	*30.5%*	*17.7%*
% margin	*32.6%*	*29.0%*	*30.6%*	*33.0%*	*34.9%*
Operating Expenses:	**(24,042)**	**(24,958)**	**(27,024)**	**(29,348)**	**(31,563)**
R&D	(15,616)	(17,725)	(18,481)	(18,900)	(19,860)
Sales and Marketing	(2,701)	(2,433)	(3,063)	(3,698)	(4,013)
General and Admin	(5,724)	(4,800)	(5,480)	(6,750)	(7,690)
Operating Profit	**51,613**	**43,949**	**55,860**	**78,823**	**95,748**
YoY Growth	*-10.8%*	*-14.8%*	*27.1%*	*41.1%*	*21.5%*
% margin	*22.2%*	*18.5%*	*20.6%*	*24.1%*	*26.2%*
Total Non-op	**4,607**	**5,699**	**9,651**	**7,791**	**9,791**
Pretax Profit	56,220	49,648	65,511	86,614	105,539
YoY Growth	*-20.7%*	*-11.7%*	*32.0%*	*32.2%*	*21.8%*
% margin	*24.2%*	*20.9%*	*24.2%*	*26.4%*	*28.9%*
Tax	(9,113)	(8,113)	(9,323)	(15,591)	(18,997)
Reported Net Income	47,211	41,716	56,406	71,242	86,760
Reported EPS (NT$)	**3.76**	**3.32**	**4.49**	**5.67**	**6.91**
ModelWare EPS (NT$)	**3.76**	**3.32**	**4.49**	**5.67**	**6.91**

Key Ratios

	2024	2025	2026e	2027e	2028e
Return (%)					
ROAA	*8.4%*	*7.3%*	*9.5%*	*11.4%*	*13.2%*
ROAE	*12.8%*	*11.0%*	*14.2%*	*16.6%*	*18.9%*
OP. ATO	*53.3%*	*54.5%*	*56.9%*	*67.4%*	*74.3%*
Gearing (x)					
Net Debt/ Equity	(0.1x)	(0.2x)	(0.2x)	(0.2x)	(0.3x)
Current Ratio	2.5x	2.3x	2.8x	3.3x	3.8x
Quick Ratio	1.8x	1.6x	1.9x	2.3x	2.7x
Operating Cycle					
AR/NR Turnover (days)	49	49	48	49	51
Inventory Turnover (days)	84	79	78	77	80
AP Turnover (days)	17	18	18	18	18
Cash Conversion (days)	115	110	108	109	112

Balance sheet

(NT$ mn)	2024	2025	2026e	2027e	2028e
Assets					
Cash & Equivalent	105,000	110,660	111,689	139,687	172,326
MarketableSecurity	606	569	528	528	528
A/R & N/R	33,343	31,274	41,114	48,236	54,317
Inventories	36,408	37,934	43,913	50,471	54,794
Other Current Ass.	11,983	20,850	29,053	29,053	29,053
Total current assets	**189,678**	**204,783**	**229,270**	**270,950**	**313,993**
LT Investment	72,487	75,373	89,312	93,312	97,312
Total Fixed Assets	279,059	271,395	262,066	246,167	232,664
Total Other Assets	28,976	27,445	28,782	28,782	28,782
Total Assets	**570,201**	**578,996**	**609,430**	**639,211**	**672,751**
A/P & N/P	7,633	9,170	10,120	11,631	12,627
Accrued Expenses	281	229	307	307	307
Other Payable	34,626	36,128	30,821	30,821	30,821
Total Current Liab.	**75,260**	**87,598**	**80,658**	**82,169**	**83,166**
L-T Liabilities	55,533	45,372	47,165	47,165	47,165
Total Other L-T Liab	61,222	66,170	66,419	66,419	66,419
Total Liabilities	**192,016**	**199,141**	**194,242**	**195,754**	**196,750**
Common Stocks	125,607	125,882	125,785	125,785	125,785
Preferred Stocks					
Capital Reserve	14,782	15,410	15,812	15,812	15,812
Retained earnings	237,539	238,477	273,555	301,824	334,368
Treasury Stock	0	0	0	0	0
Total Equity	**378,185**	**379,855**	**415,188**	**443,457**	**476,001**

Cash Flow Statement

(NT$ mn)	2024	2025	2026e	2027e	2028e
Net Income, continued ops	47,106	41,535	56,188	71,024	86,542
Depreciation	48,168	59,259	66,961	73,558	73,177
Net Investment Losses (Gains)	(72)	(99)	(11)	0	0
Others	(3,426)	(2,954)	(21,736)	(16,170)	(13,408)
Cash Flow-Operating	**93,872**	**99,864**	**101,413**	**128,412**	**146,311**
(Purchase) of FA	(88,544)	(47,745)	(57,668)	(57,659)	(59,673)
Sale of Fix Asset	121	129	10	0	0
(Purchase)L-T Inv.	(541)	0	(884)	0	0
Sale of L-T Inv.	1,602	575	79	0	0
Sale(Pur.)S-T Inv.	1,564	(8,245)	(7,301)	0	0
Cash Flow-Investing	**(85,941)**	**(53,154)**	**(66,490)**	**(57,659)**	**(59,673)**
Inc(Dec)-S-T Debt	13,178	(21,689)	1,782	0	0
Dividend Paid	(37,586)	(35,784)	(31,620)	(42,754)	(53,999)
Dir.&Emp.Bonus	0	0	0	0	0
Proceed from New Issue	0	0	0	0	0
Dec/Inc-TreasureSt	0	0	0	0	0
Others	(14,793)	18,270	(5,468)	0	0
Cash Flow-Financing	**(39,200)**	**(39,204)**	**(35,306)**	**(42,754)**	**(53,999)**
Change in Cash	**(27,553)**	**5,660**	**1,028**	**27,999**	**32,639**
Net cash/(debt), b/f	132,553	105,000	110,660	111,688	139,687
Net cash/(debt), c/f	**105,000**	**110,660**	**111,688**	**139,687**	**172,326**

e = Morgan Stanley Research Estimates
Source: Company Data, Morgan Stanley Research

Vanguard: Price target raised to NT$180 from NT$148

We raise our 2026, 2027, and 2028 earnings estimates 6%, 10%, and 8%, respectively: We see stronger data center-related demand for the overall semi industry, leading to better revenue, margins, and earnings for Vanguard as well. Besides, since TSMC will be outsourcing interposer production to Vanguard, we believe margins for the interposer business could also be better than other for power-related products in its Singapore fab.

Exhibit 15: Vanguard: Earnings estimate revisions

(NT$ mn)	New 2026e	Old 2026e	Diff.%	New 2027e	Old 2027e	Diff.%	New 2028e	Old 2028e	Diff.%
Net sales	56,799	56,799	0%	68,133	67,295	1%	77,468	75,451	3%
Gross profit	18,011	17,341	4%	21,152	19,713	7%	24,892	23,496	6%
Operating profit	11,033	10,363	6%	14,227	12,798	11%	17,862	16,480	8%
Pretax income	12,441	11,721	6%	15,723	14,297	10%	19,359	17,979	8%
Net income	10,167	9,591	6%	12,579	11,438	10%	15,487	14,383	8%
EPS (NT$)	**5.46**	**5.15**	**6%**	**6.76**	**6.14**	**10%**	**8.32**	**7.73**	**8%**
Margins									
Gross margin	31.7%	30.5%		31.0%	29.3%		32.1%	31.1%	
Operating margin	19.4%	18.2%		20.9%	19.0%		23.1%	21.8%	
Pretax margin	21.9%	20.6%		23.1%	21.2%		25.0%	23.8%	
Net margin	17.9%	16.9%		18.5%	17.0%		20.0%	19.1%	

Source: Morgan Stanley Research estimates

Exhibit 16: Vanguard: Quarterly financials

(NT$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e	2024	2025	2026e	2027e	2028e
Total Revenues	**11,949**	**11,699**	**12,349**	**12,594**	**12,532**	**14,415**	**14,770**	**15,082**	**44,055**	**48,591**	**56,799**	**68,133**	**77,468**
Sequential Change	*3.4%*	*-2.1%*	*5.6%*	*2.0%*	*-0.5%*	*15.0%*	*2.5%*	*2.1%*					
Change vs Year Ago	*24.0%*	*5.7%*	*4.6%*	*9.0%*	*4.9%*	*23.2%*	*19.6%*	*19.8%*	*15.1%*	*10.3%*	*16.9%*	*20.0%*	*13.7%*
Cost of Sales	(8,352)	(8,419)	(9,037)	(9,129)	(8,861)	(9,779)	(9,958)	(10,191)	(32,122)	(34,937)	(38,789)	(46,981)	(52,576)
Percent of Revenues	*69.9%*	*72.0%*	*73.2%*	*72.5%*	*70.7%*	*67.8%*	*67.4%*	*67.6%*	*72.9%*	*71.9%*	*68.3%*	*69.0%*	*67.9%*
Gross Profit	**3,597**	**3,280**	**3,312**	**3,465**	**3,671**	**4,637**	**4,812**	**4,891**	**11,933**	**13,654**	**18,011**	**21,152**	**24,892**
Percent of Revenues	*30.1%*	*28.0%*	*26.8%*	*27.5%*	*29.3%*	*32.2%*	*32.6%*	*32.4%*	*27.1%*	*28.1%*	*31.7%*	*31.0%*	*32.1%*
Total Opex	(1,359)	(1,384)	(1,480)	(1,658)	(1,584)	(1,708)	(1,813)	(1,872)	(4,822)	(5,880)	(6,978)	(6,926)	(7,030)
Percent of Revenues	*11.4%*	*11.8%*	*12.0%*	*13.2%*	*12.6%*	*11.9%*	*12.3%*	*12.4%*	*10.9%*	*12.1%*	*12.3%*	*10.2%*	*9.1%*
R&D	(521)	(522)	(565)	(582)	(610)	(660)	(710)	(760)	(2,081)	(2,190)	(2,741)	(2,900)	(2,900)
Percent of Revenues	*4.4%*	*4.5%*	*4.6%*	*4.6%*	*4.9%*	*4.6%*	*4.8%*	*5.0%*	*4.7%*	*4.5%*	*4.8%*	*4.3%*	*3.7%*
Sales and Marketing	(129)	(120)	(144)	(163)	(159)	(183)	(188)	(192)	(449)	(557)	(722)	(886)	(930)
Percent of Revenues	*1.1%*	*1.0%*	*1.2%*	*1.3%*	*1.3%*	*1.3%*	*1.3%*	*1.3%*	*1.0%*	*1.1%*	*1.3%*	*1.3%*	*1.2%*
General and Admin	(709)	(741)	(771)	(912)	(815)	(865)	(915)	(920)	(2,291)	(3,133)	(3,515)	(3,140)	(3,200)
Percent of Revenues	*5.9%*	*6.3%*	*6.2%*	*7.2%*	*6.5%*	*6.0%*	*6.2%*	*6.1%*	*5.2%*	*6.4%*	*6.2%*	*4.6%*	*4.1%*
Operating Income	**2,238**	**1,897**	**1,832**	**1,807**	**2,087**	**2,928**	**2,999**	**3,019**	**7,111**	**7,773**	**11,033**	**14,227**	**17,862**
Percent of Revenues	*18.7%*	*16.2%*	*14.8%*	*14.3%*	*16.7%*	*20.3%*	*20.3%*	*20.0%*	*16.1%*	*16.0%*	*19.4%*	*20.9%*	*23.1%*
Total Non-operating Income(Loss)	437	485	267	288	350	344	356	358	1,641	1,477	1,408	1,497	1,497
Profit Before Taxes	**2,675**	**2,382**	**2,099**	**2,095**	**2,437**	**3,273**	**3,355**	**3,377**	**8,752**	**9,251**	**12,441**	**15,723**	**19,359**
Percent of Revenues	*22.4%*	*20.4%*	*17.0%*	*16.6%*	*19.4%*	*22.7%*	*22.7%*	*22.4%*	*19.9%*	*19.0%*	*21.9%*	*23.1%*	*25.0%*
Change vs Year Ago	*67.7%*	*11.0%*	*-21.6%*	*-10.2%*	*-8.9%*	*37.4%*	*59.8%*	*61.2%*					
Taxes	(269)	(369)	(414)	(429)	(273)	(655)	(671)	(675)	(1,705)	(1,480)	(2,274)	(3,145)	(3,872)
Tax Rate	*10.0%*	*15.5%*	*19.7%*	*20.5%*	*11.2%*	*20.0%*	*20.0%*	*20.0%*	*19.5%*	*16.0%*	*18.3%*	*20.0%*	*20.0%*
Reported Income (TW GAAP)	**2,406**	**2,013**	**1,685**	**1,666**	**2,164**	**2,618**	**2,684**	**2,701**	**7,047**	**7,770**	**10,167**	**12,579**	**15,487**
Percent of Revenues	*20.1%*	*17.2%*	*13.6%*	*13.2%*	*17.3%*	*18.2%*	*18.2%*	*17.9%*	*16.0%*	*16.0%*	*17.9%*	*18.5%*	*20.0%*
Change vs Year Ago	*89.1%*	*11.9%*	*-21.0%*	*-9.6%*	*-10.1%*	*30.1%*	*59.2%*	*62.1%*	*-4.4%*	*10.3%*	*30.8%*	*23.7%*	*23.1%*
EPS (NT$)	**1.30**	**1.09**	**0.91**	**0.90**	**1.16**	**1.41**	**1.44**	**1.45**	**3.81**	**4.18**	**5.46**	**6.76**	**8.32**
Change vs Year Ago	*68.2%*	*-0.3%*	*-29.9%*	*-10.1%*	*-10.3%*	*29.5%*	*59.2%*	*62.1%*	*-14.5%*	*9.9%*	*30.6%*	*23.7%*	*23.1%*
Weighted Average Shares	1,857	1,853	1,862	1,862	1,862	1,862	1,862	1,862	1,852	1,858	1,862	1,862	1,862

Source: Company Data, Morgan Stanley Research estimates

We raise our price target (base case value) to NT$180 from NT$148: We factor in our earnings estimate changes for 2026-2028 and raise our intermediate growth rate from

12% to 13.5%, reflecting market share gain and overall industry demand recovery.

We continue to derive our price target from a residual income model, which we believe best captures the stock's long-term value. Besides intermediate growth rate, our other key assumptions are unchanged, including a cost of equity of 9.2% (beta of 1.2, equity risk premium of 6% and risk-free rate of 2%), and a terminal growth rate of 4.0%.

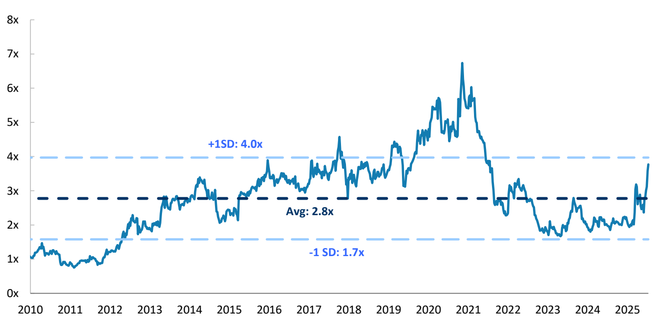
We also raise our bull and bear case values from NT$215 and NT$90 to NT$260 and NT $110, accordingly, implying 5.3x and 2.2x our 2027e BVPS.

Exhibit 17: Vanguard: Residual income model

(NT$ mn)	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e
Total Equity	85,598	90,551	96,604	107,411	119,677	133,599	149,400	167,334	187,690	210,793	237,015	266,778
Net Profit	10,167	12,579	15,487	17,578	19,951	22,644	25,701	29,171	33,109	37,579	42,652	48,410
ROAE	12.0%	14.3%	16.6%	17.2%	17.6%	17.9%	18.2%	18.4%	18.7%	18.9%	19.0%	19.2%
Residual Income	2,356	4,350	6,656	7,759	8,991	10,389	11,975	13,774	15,816	18,133	20,761	23,745
Spread	2.8%	5.1%	7.4%	8.0%	8.4%	8.7%	9.0%	9.2%	9.5%	9.7%	9.8%	10.0%
Ending Equity Capital	85,598											
PV of Forecast Period	67,684											
PV of Continuing Value	181,830											
Equity Value	335,112											
No. of Shares	1,862											
Projected Price (NT$)	180.0											

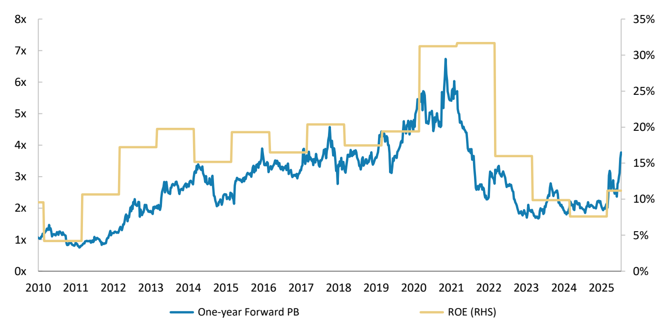
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 18: Vanguard: One-year forward P/B chart



Source: FactSet, company data, Morgan Stanley Research estimates

Exhibit 19: Vanguard: One-year forward P/B vs. ROE



Source: FactSet, company data, Morgan Stanley Research estimates

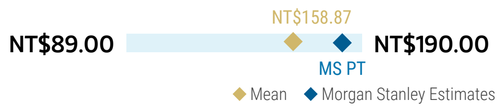
Risk Reward – Vanguard International Semiconductor (5347.TWO)

Beneficiary of overall semi industry recovery; but relative EW

PRICE TARGET NT$180.00

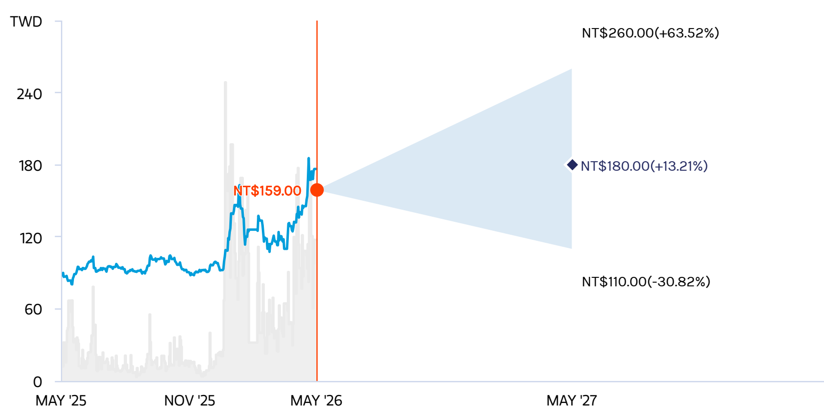
Base case, residual income model. Key assumptions: a cost of equity of 9.2%, a medium-term growth rate of 13.5%, and a terminal growth rate of 4.0%.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



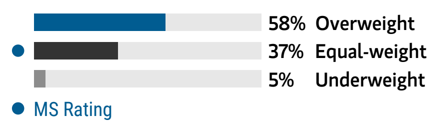
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- We expect 2026 capex to increase beyond 2025's high level given Singapore fab expansion, leading to long-term margin pressure.
- We expect PMIC wafer price hikes in the industry in response to strong demand from AI. Vanguard could benefit from TSMC mature node consolidation as well as interposer production.
- Wafer price hikes have started from 1H26, given high utilization rate..
- VSMC's 12-inch capacity is all fully booked; production will start to ramp in 2027 with depreciation pressure during the early stage.
- Our EW rating on most mature node foundries is a relative call vs. purer AI proxies like TSMC, MediaTek, SMIC, etc.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Disruption:	*Negative*
Pricing Power:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$260.00

5.3x base case 2027e BVPS

Healthy semi demand: Auto semi demand is exceptionally strong; inventory is digested quickly, and DDI for smartphones and large panels shows robust growth. We forecast a >20% revenue CAGR in 2025-28. Competition from peers eases. We assume: 1) global semi sales growth of >20% and foundry industry growth of >15% in 2026; 2) better-than-expected demand for 8" capacity; and 3) power demand for 5G tracking ahead of schedule.

BASE CASE NT$180.00

3.7x base case 2027e BVPS

Semi growth slowdown: Demand has started to pick up for LDDIC and PMIC, while auto semis inventory digestion is reaching an end. We forecast a 17% revenue CAGR in 2025-28. Vanguard starts to reverse pricing discounts to customers in view of stronger demand and some rush orders in the short run. We assume: 1) global semi sales show growth in 2026; and 2) the foundry industry shows single-digit growth in 2026.

BEAR CASE NT$110.00

2.2x base case 2027e BVPS

Semi growth decelerates: Auto semi demand weakens markedly; power semi inventory levels remain elevated. Smartphone DDI shows a worse-than-expected decline. Intensified competition and new Chinese capacity dampen the overall pricing environment. We forecast <10% revenue CAGR in 2025-28. We assume: 1) overall semi industry continues to decelerate in 2026 and foundry growth is stagnant; and 2) demand for 8" capacity is weaker than expected owing to faster product migration.

Risk Reward – Vanguard International Semiconductor (5347.TWO)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Wafer shipment (k 8 inch)	2,522	2,850	3,166	802
Wafer ASP (USD)	573.0	621.8	646.9	702.0
Utilization rate (%) (%)	71.8	80.4	81.4	78.2

INVESTMENT DRIVERS

- Smartphone/TV demand
- Persistent drop in analog IDMs' inventory
- Easing competition in 8" foundry business with pricing rebound

GLOBAL REVENUE EXPOSURE

- 0-10% Japan
- 10-20% Europe ex UK
- 10-20% Mainland China
- 10-20% North America
- 40-50% APAC, ex Japan, Mainland China and India

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

3/5 MOST | 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger-than-expected LCD driver IC demand
- Loosened competition in power semi business
- Faster-than-expected EV penetration
- Inability of IDM customers to move production in-house

RISKS TO DOWNSIDE

- Inability to fully utilize VSMC 12-inch investment
- Weaker-than-expected LCD driver IC demand
- Intensified competition in power semi business
- Slower-than-expected EV penetration
- IDM customers moving production in-house

OWNERSHIP POSITIONING

Inst. Owners, % Active 87.4%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

	Low	Morgan Stanley Estimates	Mean	High
Sales / Revenue (NT$, mn)	55,581	56,799	57,805	60,227
EBITDA (NT$, mn)	15,621	20,717	20,639	22,765
Net income (NT$, mn)	8,019	10,167	10,543	11,570
EPS (NT$)	4.34	5.46	5.64	6.20

Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Vanguard: Financial summary

Exhibit 20: Vanguard: Financial summary

Income Statement NT$ mn (Years End Dec)	2024	2025	2026e	2027e	2028e
Net sales	**44,055**	**48,591**	**56,799**	**68,133**	**77,468**
COGS	(32,122)	(34,937)	(38,789)	(46,981)	(52,576)
Gross profit	11,933	13,654	18,011	21,152	24,892
Operating expenses	(4,822)	(5,880)	(6,978)	(6,926)	(7,030)
R&D	(2,081)	(2,190)	(2,741)	(2,900)	(2,900)
SG&A	(2,740)	(3,691)	(4,237)	(4,026)	(4,130)
Operating income	**7,111**	**7,773**	**11,033**	**14,227**	**17,862**
Non-operating income	1,641	1,477	1,408	1,497	1,497
Pre-tax income	8,752	9,251	12,441	15,723	19,359
Income tax	(1,705)	(1,480)	(2,274)	(3,145)	(3,872)
Net income	**7,047**	**7,770**	**10,167**	**12,579**	**15,487**
Adj.wtd.avg.shrs (mn)	1,852	1,858	1,862	1,862	1,862
Reported EPS (NT$)	**3.81**	**4.18**	**5.46**	**6.76**	**8.32**
EPS for consensus (NT$)	**3.81**	**4.18**	**5.46**	**6.76**	**8.32**

Cash Flow Statement NT$mn (Years End Dec)	2024	2025	2026e	2027e	2028e
Cash flow from Operations	**24,731**	**25,439**	**(1,042)**	**22,554**	**26,565**
Net profits	7,047	7,770	10,167	12,579	15,487
Depreciation	8,622	8,552	9,684	10,796	11,851
Equity investment losses (income)	(84)	(209)	0	0	0
Other adjustments	9,145	9,326	(20,893)	(821)	(774)
Cash flow from Investing	**(11,699)**	**(59,711)**	**(63,781)**	**(26,275)**	**(25,826)**
Capex	(15,909)	(63,265)	(63,781)	(26,275)	(25,826)
Change of LT Investment	(2,480)	(0)	0	0	0
Change of ST Investment	6,837	1,183	0	0	0
Other adjustments	(147)	2,371	0	0	0
Cash flow from financing	**17,530**	**13,460**	**71,841**	**(7,625)**	**(9,434)**
Increase in L/T debt	0	2,000	0	0	0
Increase in S/T debt	0	5,519	80,000	0	0
Cash Dividend Paid	(7,375)	(8,400)	(8,159)	(7,625)	(9,434)
Dir& Emp Bonus Paid	0	0	0	0	0
Issuance of stock	0	0	0	0	0
Dec/Inc-TreasureSt	0	0	0	0	0
Other Adjustments	21,905	14,341	0	0	0
Net change in cash	**30,562**	**(20,812)**	**7,019**	**(11,346)**	**(8,696)**

Balance Sheet NT$mn (Years End Dec)	2024	2025	2026e	2027e	2028e
Cash	60,695	39,635	46,654	35,307	26,612
Mkt Securities	8,929	2,640	2,640	2,640	2,640
AR/NR	6,235	6,748	7,588	9,102	10,349
Inventory	6,221	6,937	7,304	8,847	9,900
Other	4,131	10,937	10,937	10,937	10,937
Current Assets	**86,211**	**66,897**	**75,123**	**66,833**	**60,438**
Long-term investments	7,448	9,420	9,420	9,420	9,420
Fixed assets	44,141	113,869	167,966	183,445	197,420
Other assets	10,905	10,282	10,282	10,282	10,282
Total Assets	**148,706**	**200,469**	**262,791**	**269,980**	**277,560**
S/T borrowings	0	5,521	5,521	5,521	5,521
AP/NP	1,416	1,724	1,442	1,747	1,955
Other ST liabilities	19,959	44,359	104,955	106,887	108,206
LT debt	20,984	15,990	15,990	15,990	15,990
Other LT liabilities	37,679	49,285	49,285	49,285	49,285
Total Liabilities	**80,039**	**116,879**	**177,193**	**179,429**	**180,956**
Common shares	18,666	18,671	18,671	18,671	18,671
Retained earning	31,135	45,904	47,912	52,865	58,918
Other shareholders' equity	18,865	19,015	19,015	19,015	19,015
Total Equity	**68,667**	**83,590**	**85,598**	**90,551**	**96,604**
Total Liab. & Shrhldr's Equity	**148,706**	**200,469**	**262,791**	**269,980**	**277,560**

Financial Ratios	2024	2025	2026e	2027e	2028e
Growth(%)					
Turnover	15.1	10.3	16.9	20.0	13.7
Operating profits	23.2	9.3	41.9	28.9	25.6
Pretax profits	(0.9)	5.7	34.5	26.4	23.1
Net profits	(4.4)	10.3	30.8	23.7	23.1
EPS	(14.5)	9.9	30.6	23.7	23.1
Margins (%)					
Gross Margin	27.1	28.1	31.7	31.0	32.1
Operating Margin	16.1	16.0	19.4	20.9	23.1
Pretax Margin	19.9	19.0	21.9	23.1	25.0
Net Profit	16.0	16.0	17.9	18.5	20.0
Return (%)					
ROAA	5.6	4.5	4.4	4.7	5.7
ROAE	12.3	10.2	12.0	14.3	16.6
Gearing (%)					
Net Debt/Equity	(57.8)	(21.7)	(29.4)	(15.2)	(5.3)
Liabilities/Equity	116.6	139.8	207.0	198.2	187.3
Ratios (X)					
Current ratio	4.0	1.3	0.7	0.6	0.5
Quick ratio	3.5	0.9	0.5	0.4	0.3
Others					
AR/NR Turnover (days)	48.7	48.8	48.8	48.8	48.8
Inventory Turnover (days)	67.6	68.7	68.7	68.7	68.7
AP Turnover (days)	13.6	13.6	13.6	13.6	13.6
Cash Conversion (days)	102.7	103.9	103.9	103.9	103.9

E = Morgan Stanley Research Estimates
Source: Morgan Stanley Research, Company Data

GlobalWafers: Downgrade from OW to EW; price target raised to NT$750

We downgrade GWC from Overweight to Equal-weight: We update our model with 1Q26 preliminary results. We also raise our 2028 shipment forecast – the company has mentioned that 2027 and 2028 could show even stronger growth than 2026 given the industry recovery.

The 2H26 pricing recovery, improving LTA fulfillment, and compound semiconductor progress are constructive. However, we believe much of this is increasingly reflected in expectations. At the same time, higher production costs and new capacity ramp could limit the pace of margin recovery. With pricing upside back-end-loaded and cost pressure still visible, we see a more balanced setup from here.

Exhibit 21: GWC: Estimate revisions

(NT$ mn)	New 2026e	Old 2026e	Diff.%	New 2027e	Old 2027e	Diff.%	New 2028e	Old 2028e	Diff.%
Net sales	69,107	70,755	-2%	85,788	88,242	-3%	98,141	99,360	-1%
Gross profit	17,360	20,439	-15%	24,643	25,759	-4%	31,237	30,368	3%
Operating profit	11,199	14,578	-23%	18,108	19,181	-6%	23,905	23,189	3%
Pretax income	14,065	15,897	-12%	21,305	22,099	-4%	27,094	26,108	4%
Net income	10,684	11,922	-10%	15,979	16,574	-4%	20,320	19,581	4%
EPS for Consensus	**22.34**	**24.92**	**-10%**	**33.40**	**34.65**	**-4%**	**42.48**	**40.93**	**4%**
Margins									
Gross margin	25.1%	28.9%		28.7%	29.2%		31.8%	30.6%	
Operating margin	16.2%	20.6%		21.1%	21.7%		24.4%	23.3%	
Pretax margin	20.4%	22.5%		24.8%	25.0%		27.6%	26.3%	
Net margin	15.5%	16.8%		18.6%	18.8%		20.7%	19.7%	

Source: Morgan Stanley Research (e) estimates

Exhibit 22: GWC: Quarterly financials

(NT$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26e	2Q26e	3Q26e	4Q26e	2024	2025	2026e	2027e	2028e
Total Revenues	**15,595**	**16,008**	**14,493**	**14,502**	**13,985**	**15,810**	**18,346**	**20,967**	**62,626**	**60,598**	**69,107**	**85,788**	**98,141**
Sequential Change	*-4.6%*	*2.7%*	*-9.5%*	*0.1%*	*-3.6%*	*13.0%*	*16.0%*	*14.3%*					
Change vs Year Ago	*3.4%*	*4.5%*	*-8.7%*	*-11.3%*	*-10.3%*	*-1.2%*	*26.6%*	*44.6%*	*-11.4%*	*-3.2%*	*14.0%*	*24.1%*	*14.4%*
Cost of Sales	**(11,483)**	**(11,885)**	**(11,831)**	**(10,776)**	**(11,071)**	**(11,996)**	**(13,467)**	**(15,213)**	**(42,823)**	**(45,974)**	**(51,747)**	**(61,145)**	**(66,904)**
Gross Profit	**4,112**	**4,123**	**2,662**	**3,726**	**2,914**	**3,814**	**4,879**	**5,754**	**19,804**	**14,624**	**17,360**	**24,643**	**31,237**
Gross Margin	26.4%	25.8%	18.4%	25.7%	20.8%	24.1%	26.6%	27.4%	31.6%	24.1%	25.1%	28.7%	31.8%
Total Opex	(1,523)	(1,685)	(1,433)	(1,347)	(1,438)	(1,497)	(1,575)	(1,651)	(5,685)	(5,987)	(6,161)	(6,534)	(7,332)
Percent of Revenues	*9.8%*	*10.5%*	*9.9%*	*9.3%*	*10.3%*	*9.5%*	*8.6%*	*7.9%*	*9.1%*	*9.9%*	*8.9%*	*7.6%*	*7.5%*
R&D	(611)	(576)	(514)	(516)	(471)	(476)	(481)	(486)	(2,363)	(2,218)	(1,914)	(2,445)	(2,820)
Percent of Revenues	*3.9%*	*3.6%*	*3.5%*	*3.6%*	*3.4%*	*3.0%*	*2.6%*	*2.3%*	*3.8%*	*3.7%*	*2.8%*	*2.9%*	*2.9%*
Sales and Marketing	(383)	(496)	(360)	(361)	(375)	(424)	(492)	(563)	(1,352)	(1,600)	(1,855)	(1,544)	(1,767)
Percent of Revenues	*2.5%*	*3.1%*	*2.5%*	*2.5%*	*2.7%*	*2.7%*	*2.7%*	*2.7%*	*2.2%*	*2.6%*	*2.7%*	*1.8%*	*1.8%*
General and Admin	(528)	(613)	(559)	(469)	(592)	(597)	(602)	(602)	(1,970)	(2,169)	(2,391)	(2,545)	(2,745)
Percent of Revenues	*3.4%*	*3.8%*	*3.9%*	*3.2%*	*4.2%*	*3.8%*	*3.3%*	*2.9%*	*3.1%*	*3.6%*	*3.5%*	*3.0%*	*2.8%*
Operating Income	**2,589**	**2,438**	**1,230**	**2,379**	**1,475**	**2,317**	**3,304**	**4,103**	**14,118**	**8,636**	**11,199**	**18,108**	**23,905**
Percent of Revenues	*16.6%*	*15.2%*	*8.5%*	*16.4%*	*10.5%*	*14.7%*	*18.0%*	*19.6%*	*22.5%*	*14.3%*	*16.2%*	*21.1%*	*24.4%*
Total Non-operating Income(Loss)	**(456)**	**(150)**	**957**	**528**	**871**	**620**	**659**	**716**	**(1,690)**	**880**	**2,866**	**3,197**	**3,188**
Profit Before Taxes	**2,133**	**2,289**	**2,187**	**2,907**	**2,347**	**2,936**	**3,963**	**4,819**	**12,429**	**9,516**	**14,065**	**21,305**	**27,094**
Percent of Revenues	*13.7%*	*14.3%*	*15.1%*	*20.0%*	*16.8%*	*18.6%*	*21.6%*	*23.0%*	*19.8%*	*15.7%*	*20.4%*	*24.8%*	*27.6%*
Change vs Year Ago	*-53.2%*	*-35.2%*	*-38.3%*	*265.2%*	*10.0%*	*28.3%*	*81.2%*	*65.8%*					
Taxes	(678)	(607)	(218)	(702)	(451)	(734)	(991)	(1,205)	(2,590)	(2,205)	(3,380)	(5,326)	(6,773)
Tax Rate	*31.8%*	*26.5%*	*10.0%*	*24.2%*	*19.2%*	*25.0%*	*25.0%*	*25.0%*	*20.8%*	*23.2%*	*24.0%*	*25.0%*	*25.0%*
Net Income, Cont Ops	1,456	1,682	1,969	2,205	1,896	2,202	2,972	3,614	9,839	7,312	10,684	15,979	20,320
Percent of Revenues	*9.3%*	*10.5%*	*13.6%*	*15.2%*	*13.6%*	*13.9%*	*16.2%*	*17.2%*	*15.7%*	*12.1%*	*15.5%*	*18.6%*	*20.7%*
Minority Interest	1	(0)	(0)	(0)	(0)	(0)	(0)	(0)	7	0	(0)	(0)	(0)
Reported Income (TW GAAP)	**1,456**	**1,682**	**1,969**	**2,205**	**1,896**	**2,202**	**2,972**	**3,614**	**9,846**	**7,312**	**10,684**	**15,979**	**20,320**
Percent of Revenues	*9.3%*	*10.5%*	*13.6%*	*15.2%*	*13.6%*	*13.9%*	*16.2%*	*17.2%*	*15.7%*	*12.1%*	*15.5%*	*18.6%*	*20.7%*
Change vs Year Ago	*-58.8%*	*-41.6%*	*-33.3%*	*360.5%*	*30.2%*	*30.9%*	*50.9%*	*63.9%*	*-50.2%*	*-25.7%*	*46.1%*	*49.6%*	*27.2%*
Reported Diluted EPS (NT$, TW GAAP)	**3.04**	**3.51**	**4.12**	**4.61**	**3.96**	**4.60**	**6.21**	**7.56**	**21.14**	**15.28**	**22.34**	**33.40**	**42.48**
Reported Basic EPS (NT$, GAAP)	**3.05**	**3.52**	**4.12**	**4.61**	**3.97**	**4.61**	**6.22**	**7.56**	**21.06**	**15.29**	**22.35**	**33.42**	**42.50**

Source: Company data, Morgan Stanley Research (e) estimates

We raise our price target (base case value) to NT$750: We continue to derive our price target from a residual income model because we believe it best presents the value of the company. We are raising our intermediate growth rate from 11.5% to 13%; our other key assumptions are unchanged, including a cost of equity of 11% (beta of 1.5, equity risk premium of 6%, and risk-free rate of 2%) and terminal growth rate of 4%.

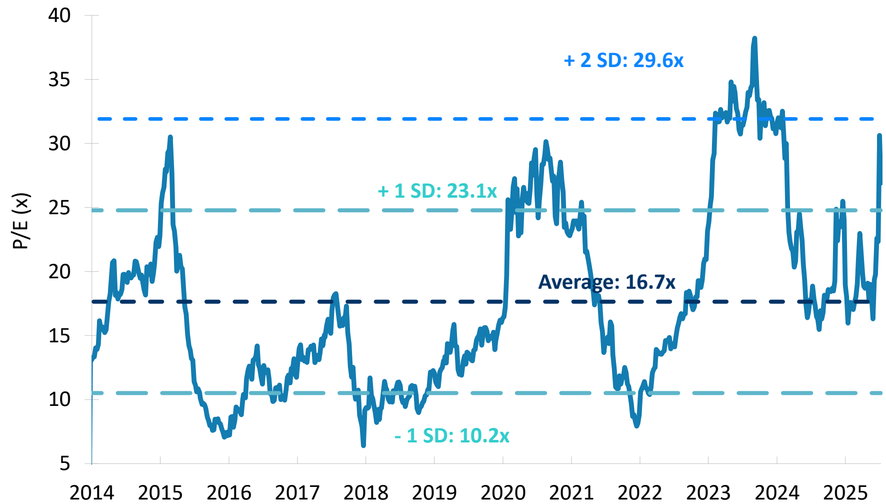
Our bull case value is also raised to NT$950 from NT$750 and our bear case value to NT $380 from NT$300 , implying 29x and 11x 2027e EPS, respectively.

Exhibit 23: GWC: Residual income model

(NT$ mn)	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e
Total Equity	**99,592**	**109,160**	**119,893**	**125,764**	**132,399**	**139,896**	**148,368**	**157,940**	**168,758**	**180,982**	**194,794**	**210,403**
Net Profit	**10,684**	**15,979**	**20,320**	**22,962**	**25,947**	**29,320**	**33,131**	**37,438**	**42,305**	**47,805**	**54,020**	**61,042**
ROAE	**11.1%**	**15.3%**	**17.7%**	**18.7%**	**20.1%**	**21.5%**	**23.0%**	**24.4%**	**25.9%**	**27.3%**	**28.8%**	**30.1%**
Residual Income	73	4,291	7,360	9,225	11,446	13,949	16,769	19,948	23,531	27,571	32,126	37,263
Spread	**0.1%**	**4.3%**	**6.7%**	**7.7%**	**9.1%**	**10.5%**	**12.0%**	**13.4%**	**14.9%**	**16.3%**	**17.8%**	**19.1%**
Ending Equity Capital	99,592											
PV of Forecast Period	83,278											
PV of Continuing Value	175,656											
Equity Value	**358,527**											
No. of Shares	478											
Projected Price (NT$)	**750**											

Source: Morgan Stanley Research (e) estimates.

Exhibit 24: GWC's one-year forward P/B band



Source: FactSet, Morgan Stanley Research estimates

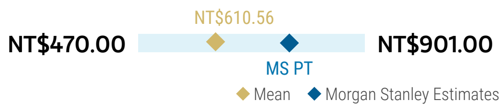
Risk Reward – GlobalWafers Co Ltd (6488.TWO)

We see a more balanced setup from here

PRICE TARGET NT$750.00

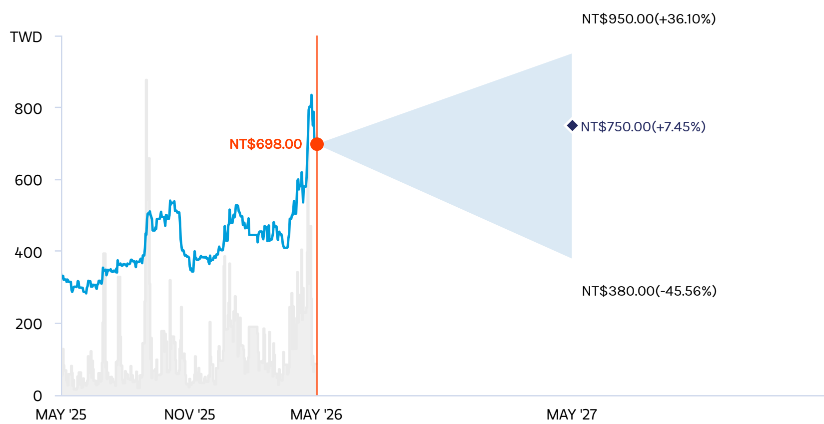
Base-case, residual income model. We assume a cost of equity of 11.0% (risk-free rate of 2.0%, beta of 1.5 and equity risk premium of 6.0%), intermediate-term growth rate of 13%, and terminal growth rate of 4.0%.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



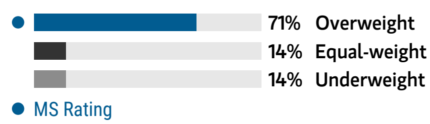
Key: Historical Stock Performance, Current Stock Price, Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- The 2H26 pricing recovery, improving LTA fulfillment, and compound semiconductor progress are constructive.
- However, we believe much of this is increasingly reflected in expectations.
- At the same time, higher production costs and new capacity ramp could limit the pace of margin recovery.
- With pricing upside back-end-loaded and cost pressure still visible, we see a more balanced setup from here.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$950.00

29x 2027e EPS

Semiconductor industry recovers faster than expected. Pure foundry players' utilization rates rebound to 100%, and memory players start to pull in more raw wafers. GWC's total revenue CAGR picks up to >20% in 2025-28 vs. 8% in 2021-23. Gross margin recovers to over 40% vs. 37.4% in 2023.

BASE CASE NT$750.00

22x 2027e EPS

Semiconductor industry recovers from a down-cycle. Pure foundry and memory players start to see recovery in utilization rate, while both spot price and LTA amount also increase gradually. GWC's total revenue CAGR rises to >15% in 2025-28 vs. 8% in 2021-23. Gross margin stays above 25%.

BEAR CASE NT$380.00

11x 2027e EPS

Semiconductor industry recovers more slowly than expected during this down-cycle. Pure foundry players start to breach LTAs and memory players start to cut more raw wafer orders. GWC's total revenue CAGR slows to 0% in 2025-28 vs. 8% in 2021-23. Gross margin declines to <20% vs. 37.4% in 2023.

Risk Reward – GlobalWafers Co Ltd (6488.TWO)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Total Wafer Shipment (NT$, 000s)	32,501	34,688	37,895	32,035
GWC Utilization Rate (%)	51.3	49.9	49.6	55.8
Blended ASP Per Wafer (NT$)	60	64	73	61

INVESTMENT DRIVERS

- Monthly sales inflection
- Stronger outlooks from key customers
- Extension of LTA pricing or shipments by GWC or peers

GLOBAL REVENUE EXPOSURE

- **10-20%** Europe ex UK
- **10-20%** North America
- **20-30%** Mainland China
- **30-40%** APAC, ex Japan, Mainland China and India

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

5/5 **MOST** | 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Margins improve as loosening competition boosts overall wafer ASP.
- Wafer supply remains stable or even goes into shortage amid stronger-than-expected wafer demand.

RISKS TO DOWNSIDE

- Global semiconductor growth slows again, triggering less wafer demand.
- Margins erode as rising competition lowers overall wafer ASP.
- Global wafer supply rises unexpectedly from new projects.

OWNERSHIP POSITIONING

Inst. Owners, % Active 60.4%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

	Low	Morgan Stanley Estimates	Mean	High
Sales / Revenue (NT$, mn)	61,936	69,107	65,895	78,314
EBITDA (NT$, mn)	16,750	22,946	21,377	26,325
Net income (NT$, mn)	7,337	10,684	10,352	17,579
EPS (NT$)	15.34	22.34	21.88	40.20

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

GWC: Financial Summary

Exhibit 25: GWC: Financial Summary

Income Statement, 2024-2028e, Year End Dec

(NT$ mn)	2024	2025	2026e	2027e	2028e
Turnover	**62,626**	**60,598**	**69,107**	**85,788**	**98,141**
YoY Growth	*-11.4%*	*-3.2%*	*14.0%*	*24.1%*	*14.4%*
Less: COGS	**(42,823)**	**(45,974)**	**(51,747)**	**(61,145)**	**(66,904)**
Variable costs	(35,056)	(37,362)	(40,442)	(46,754)	(48,089)
Depreciation & amort	(7,766)	(8,612)	(11,305)	(14,391)	(18,815)
Gross profit	**19,804**	**14,624**	**17,360**	**24,643**	**31,237**
YoY Growth	*-25.1%*	*-26.2%*	*18.7%*	*41.9%*	*26.8%*
% margin	*31.6%*	*24.1%*	*25.1%*	*28.7%*	*31.8%*
Operating Expenses:	**(5,685)**	**(5,987)**	**(6,161)**	**(6,534)**	**(7,332)**
R&D	(2,363)	(2,218)	(1,914)	(2,445)	(2,820)
Sales and Marketing	(1,352)	(1,600)	(1,855)	(1,544)	(1,767)
General and Admin	(1,970)	(2,169)	(2,391)	(2,545)	(2,745)
Operating Profit	**14,118**	**8,636**	**11,199**	**18,108**	**23,905**
YoY Growth	*-29.6%*	*-38.8%*	*29.7%*	*61.7%*	*32.0%*
% margin	*22.5%*	*14.3%*	*16.2%*	*21.1%*	*24.4%*
Total Non-op	**(1,690)**	**880**	**2,866**	**3,197**	**3,188**
Pretax Profit	12,429	9,516	14,065	21,305	27,094
YoY Growth	*-53.1%*	*-23.4%*	*47.8%*	*51.5%*	*27.2%*
% margin	*19.8%*	*15.7%*	*20.4%*	*24.8%*	*27.6%*
Tax	(2,590)	(2,205)	(3,380)	(5,326)	(6,773)
Reported Net Income	9,846	7,312	10,684	15,979	20,320
EPS (NT$)	**21.14**	**15.28**	**22.34**	**33.40**	**42.48**

Key Ratios

	2024	2025	2026e	2027e	2028e
Return (%)					
ROAA	*4.3%*	*2.5%*	*3.5%*	*4.7%*	*5.6%*
ROAE	*11.2%*	*7.1%*	*10.1%*	*14.6%*	*17.7%*
OP. ATO	*51.1%*	*38.6%*	*39.4%*	*48.5%*	*55.2%*
Gearing (x)					
Net Debt/ Equity	(0.0x)	0.3x	0.4x	0.3x	0.1x
Current Ratio	1.3x	1.6x	2.6x	3.5x	4.4x
Quick Ratio	0.8x	0.5x	1.5x	2.4x	3.2x
Operating Cycle					
AR/NR Turnover (days)	65	83	78	69	64
Inventory Turnover (days)	96	109	102	95	90
AP Turnover (days)	42	44	39	36	34
Cash Conversion (days)	118	149	140	128	120

Balance sheet

(NT$ mn)	2024	2025	2026e	2027e	2028e
Assets					
Cash & Equivalents	38,929	19,484	67,546	115,723	165,399
Marketable Security	29	1	1	1	1
A/R & N/R	10,265	10,113	14,621	15,780	19,062
Inventories	11,238	10,399	14,681	15,650	17,827
Other Current Asset	20,030	46,632	46,632	46,632	46,632
Total current assets	**81,349**	**87,237**	**144,089**	**194,394**	**249,528**
LT Investment	14,280	14,811	14,811	14,811	14,811
Total Fixed Assets	119,074	107,241	117,908	117,310	113,509
Total Other Assets	9,877	9,054	9,286	9,518	9,750
Total Assets	**224,581**	**218,343**	**286,094**	**336,033**	**387,599**
A/P & N/P	5,371	4,161	5,615	5,986	6,818
Accrued Expenses	0	0	0	0	0
Other Payable	4,650	3,613	3,613	3,613	3,613
Total Current Liab.	**65,065**	**54,109**	**55,563**	**55,934**	**56,766**
L-T Liabilities	37,678	43,244	103,244	143,244	183,244
Total Other L-T Liab	30,810	27,695	27,695	27,695	27,695
Total Liabilities	**133,553**	**125,048**	**186,502**	**226,873**	**267,706**
Common Stock	4,781	4,781	4,781	4,781	4,781
Preferred Stock					
Capital Reserve	45,720	44,764	44,764	44,764	44,764
Retained earnings	40,530	43,753	50,051	59,619	70,351
Treasury Stock	0	0	0	0	0
Total Equity	**91,028**	**93,295**	**99,592**	**109,160**	**119,893**

Cash Flow Statement

(NT$ mn)	2024	2025	2026e	2027e	2028e
Net Income	9,846	7,312	10,684	15,979	20,320
Depreciation	8,069	8,949	11,747	14,953	19,550
Net Investment Losses (Gains)	(49)	48	143	143	143
Others	(2,825)	(3,564)	(7,336)	(1,757)	(4,626)
Cash Flow-Operating	**15,041**	**12,745**	**15,238**	**29,317**	**35,387**
(Purchase) of FA	(48,319)	(33,497)	(22,556)	(14,497)	(15,892)
Sale of Fix Asset	380	86	0	0	0
(Purchase)L-T Inv.	0	0	0	0	0
Sale of L-T Inv.	0	0	0	0	0
Sale(Pur.)S-T Inv.	(6,846)	(106)	0	0	0
Cash Flow-Investing	**(29,620)**	**(34,049)**	**(22,788)**	**(14,729)**	**(16,124)**
Inc(Dec)-S-T Debt	8,222	(3,977)	0	0	0
Dividend Paid	(8,748)	(5,259)	(4,387)	(6,411)	(9,587)
Dir.&Emp.Bonus	0	0	0	0	0
Proceed from New Issue	21,891	0	0	0	0
Dec/Inc-TreasureSt	0	0	0	0	0
Others	4,375	10,540	60,000	40,000	40,000
Cash Flow-Financing	**25,740**	**1,303**	**55,613**	**33,589**	**30,413**
Change in Cash	**12,765**	**(19,445)**	**48,062**	**48,177**	**49,675**
Net cash/(debt), b/f	26,165	38,929	19,484	67,546	115,723
Net cash/(debt), c/f	**38,929**	**19,484**	**67,546**	**115,723**	**165,399**

e = Morgan Stanley Research Estimates
Source: Company Data, Morgan Stanley Research

Silergy: Estimate revisions

We cut our 2026 EPS estimate 5%... We factor in the 1Q26 margin miss from inventory valuation losses, and increasing production cost. 1Q26 GM was 48.2%, down 3.0ppts Q/Q and down 5.7ppts Y/Y – margin impacts include:

- Inventory loss from accounting method
- Auto revenue (higher margin) declined from 15% of the mix in 4Q25 to <13% in 1Q26
- Notebook revenue (lower margin) increased from <10% of the mix in 4Q25 to >12% in 1Q26 given order pull-in
- increasing production cost from foundry wafer price hike

We think the increasing production cost should still cap the margin from here. Silergy's mix still has more than 30% of consumer-related revenue, and we don't expect the company to fully pass through the rising cost given weaker end market demand.

Exhibit 26: Silergy: 1Q26 Earnings Review

(NT$ mn)	1Q26				
	Actual	Q/Q	Y/Y	MS Est.	Consensus
Revenue	4,860	-9.9%	18.7%	4,861	4,823
GM (%)	48.2%	-298bps	-568bps	52.3%	51.8%
Opex	(1,953)	-2.4%	-0.7%	(1,823)	(1,953)
OPM (%)	8.0%	-606bps	217bps	14.8%	11.3%
EPS (NT$)	1.98	-3.5%	112.4%	1.86	1.51

Source: Company Data, Visible Alpha, Morgan Stanley Research estimates

...while 2027 and 2028e estimates remain roughly unchanged.

Exhibit 27: Silergy: Estimate revision summary

NT$ mn	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
Net sales	23,695	23,697	0%	28,515	28,518	0%	33,920	33,923	0%
Gross profit	12,268	12,270	0%	15,203	15,204	0%	18,179	18,180	0%
Operating profit	3,588	4,164	-14%	6,767	6,747	0%	8,840	8,756	1%
Pretax Income	4,221	4,552	-7%	7,083	7,136	-1%	9,156	9,144	0%
Net income	4,000	4,217	-5%	6,495	6,542	-1%	8,360	8,350	0%
Basic EPS (NT$)	**10.14**	**10.69**	**-5%**	**16.47**	**16.59**	**-1%**	**21.20**	**21.18**	**0%**
Margins									
Gross margin	51.8%	51.8%		53.3%	53.3%		53.6%	53.6%	
Operating margin	15.1%	17.6%		23.7%	23.7%		26.1%	25.8%	
Pretax margin	17.8%	19.2%		24.8%	25.0%		27.0%	27.0%	
Net margin	16.9%	17.8%		22.8%	22.9%		24.6%	24.6%	

Source: Morgan Stanley Research (E) estimates

Exhibit 28: Silergy: Quarterly financials

NT$ in million	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	3Q26E	4Q26E	2024	2025	2026E	2027E	2028E
Total Revenues	4,094	4,564	4,763	5,392	4,860	5,589	6,427	6,818	18,455	18,812	23,695	28,515	33,920
Sequential Change	*-19.9%*	*11.5%*	*4.4%*	*13.2%*	*-9.9%*	*15.0%*	*15.0%*	*6.1%*					
Change vs Year Ago	*6.6%*	*-1.1%*	*-2.6%*	*5.5%*	*18.7%*	*22.5%*	*34.9%*	*26.5%*	*19.6%*	*1.9%*	*26.0%*	*20.3%*	*19.0%*
Cost of Sales	1,888	2,215	2,368	2,632	2,517	2,713	3,124	3,272	8,519	9,102	11,426	13,312	15,741
Percent of Revenues	*46%*	*49%*	*50%*	*49%*	*52%*	*49%*	*49%*	*48%*	*46%*	*48%*	*48%*	*47%*	*46%*
Gross Profit	2,206	2,349	2,395	2,760	2,343	2,876	3,304	3,546	9,936	9,710	12,268	15,203	18,179
Percent of Revenues	*53.9%*	*51.5%*	*50.3%*	*51.2%*	*48.2%*	*51.5%*	*51.4%*	*52.0%*	*53.8%*	*51.6%*	*51.8%*	*53.3%*	*53.6%*
Incremental Margin	*NM*	*30%*	*23%*	*58%*	*NM*	*73%*	*51%*	*62%*	*111%*	*-63%*	*52%*	*61%*	*55%*
Total Opex	1,966	1,809	1,791	2,001	1,953	2,071	2,204	2,452	7,682	7,567	8,681	8,435	9,338
Percent of Revenues	*48.0%*	*39.6%*	*37.6%*	*37.1%*	*40.2%*	*37.1%*	*34.3%*	*36.0%*	*41.6%*	*40.2%*	*36.6%*	*29.6%*	*27.5%*
R&D	1,329	1,223	1,213	1,346	1,367	1,467	1,567	1,667	5,151	5,111	6,067	5,660	6,000
Percent of Revenues	*32.5%*	*26.8%*	*25.5%*	*25.0%*	*28.1%*	*26.2%*	*24.4%*	*24.4%*	*27.9%*	*27.2%*	*25.6%*	*19.8%*	*17.7%*
General & administrative	283	216	217	187	231	241	251	261	911	901	986	1,100	1,340
Percent of Revenues	*6.9%*	*4.7%*	*4.6%*	*3.5%*	*4.8%*	*4.3%*	*3.9%*	*3.8%*	*4.9%*	*4.8%*	*4.2%*	*3.9%*	*4.0%*
Selling & marketing	355	370	361	468	355	363	386	524	1,621	1,554	1,628	1,675	1,998
Percent of Revenues	*8.7%*	*8.1%*	*7.6%*	*8.7%*	*7.3%*	*6.5%*	*6.0%*	*7.7%*	*8.8%*	*8.3%*	*6.9%*	*5.9%*	*5.9%*
Operating Income	240	540	604	759	390	805	1,100	1,094	2,254	2,144	3,588	6,767	8,840
Percent of Revenues	*5.9%*	*11.8%*	*12.7%*	*14.1%*	*8.0%*	*14.4%*	*17.1%*	*16.0%*	*12.2%*	*11.4%*	*15.1%*	*23.7%*	*26.1%*
Change vs Year Ago	*nm*	*nm*	*nm*	*nm*	*nm*	*nm*	*nm*	*nm*	*nm*	*nm*	*67.4%*	*88.6%*	*30.6%*
Total Non-operating Income(Loss)	140	87	200	107	397	79	79	79	329	533	634	316	316
Profit Before Taxes	380	627	804	866	787	884	1,179	1,173	2,583	2,677	4,221	7,083	9,156
Percent of Revenues	*9%*	*14%*	*17%*	*16%*	*16%*	*16%*	*18%*	*17%*	*14%*	*14%*	*18%*	*25%*	*27%*
Taxes	20	(1)	79	507	159	88	118	117	459	605	482	708	916
Tax Rate	*5.3%*	*-0.2%*	*9.8%*	*58.5%*	*20.2%*	*10.0%*	*10.0%*	*10.0%*	*17.8%*	*22.6%*	*11.4%*	*10.0%*	*10.0%*
Reported Income (TW GAAP)	**359**	**630**	**680**	**809**	**769**	**816**	**1,101**	**1,116**	**2,286**	**2,478**	**4,000**	**6,495**	**8,360**
Percent of Revenues	*9%*	*14%*	*14%*	*15%*	*16%*	*15%*	*17%*	*16%*	*12%*	*13%*	*17%*	*23%*	*25%*
Reported EPS (NT$, TW GAAP)	**0.93**	**1.64**	**1.73**	**2.05**	**1.95**	**2.07**	**2.79**	**2.83**	**5.90**	**6.36**	**10.14**	**16.47**	**21.20**
Change vs Year Ago	*372%*	*12%*	*-12%*	*-12%*	*109%*	*26%*	*62%*	*38%*	*214%*	*7%*	*52%*	*71%*	*29%*

Source: Company data, Morgan Stanley Research (e) estimates

Silergy: Valuation Methodology

Our price target remains at NT$388, but we downgrade the stock to UW: All of our key assumptions are unchanged, including a cost of equity of 9.5% (2.0% risk-free rate, 6% risk premium, beta of 1.25), intermediate growth rate of 13.5%, and terminal growth rate of 4.5%.

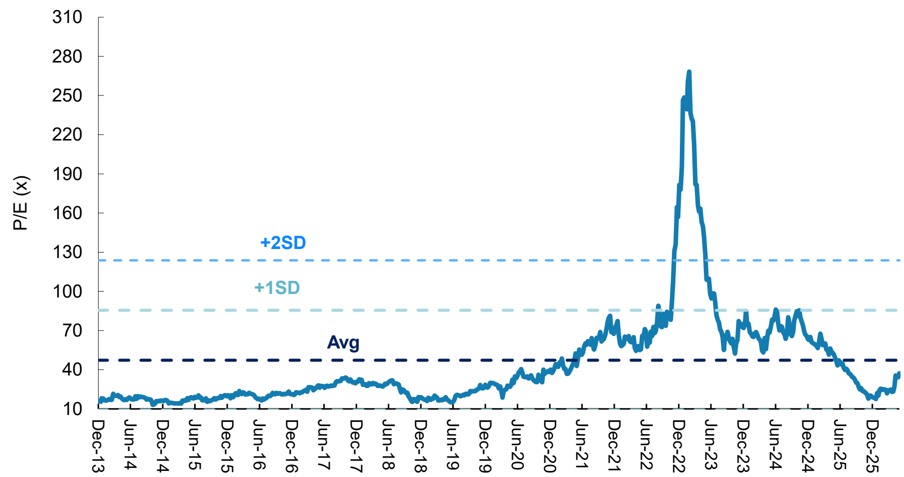
Our bull and bear scenario values also stay at NT$590 and NT$210, respectively, implying 35x and 12x 2027e EPS. However, we downgrade the stock to UW given the potential margin impact from foundry price hike and our belief that the revenue upside has been more than fully priced in.

Exhibit 10 Silergy: Residual income model

NT$million	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	**40,748**	**45,620**	**51,345**	**58,742**	**67,137**	**76,666**	**87,480**	**99,755**	**113,687**	**129,500**	**147,447**	**167,817**
Net Profit	**4,000**	**6,495**	**8,360**	**9,489**	**10,770**	**12,224**	**13,874**	**15,747**	**17,873**	**20,286**	**23,024**	**26,133**
ROAE	**10.2%**	**15.0%**	**17.2%**	**17.2%**	**17.1%**	**17.0%**	**16.9%**	**16.8%**	**16.7%**	**16.7%**	**16.6%**	**16.6%**
Residual Income	261	2,257	3,533	3,974	4,471	5,036	5,677	6,404	7,230	8,166	9,230	10,437
Spread	**0.7%**	**5.5%**	**7.7%**	**7.7%**	**7.6%**	**7.5%**	**7.4%**	**7.3%**	**7.2%**	**7.2%**	**7.1%**	**7.1%**
Ending Equity Capital	40,748											
PV of Forecast Period	31,859											
PV of Continuing Value	80,380											
Equity Value	**152,987**											
No. of Shares	394											
Projected Price	**388**											

Source: Morgan Stanley Research estimates

Exhibit 11 Silergy: Historical P/E multiples



Source: Company data, Morgan Stanley Research (e) estimates

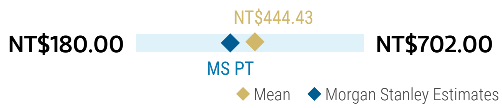
Risk Reward – Silergy Corp. (6415.TW)

Upside more than fully priced in; move to UW

PRICE TARGET NT$388.00

Base case, residual income model. We assume a cost of equity of 9.5% (2.0% risk-free rate, 6% risk premium, beta of 1.25), an intermediate growth rate of 13.5%, and a terminal growth rate of 4.5%.

Consensus Price Target Distribution

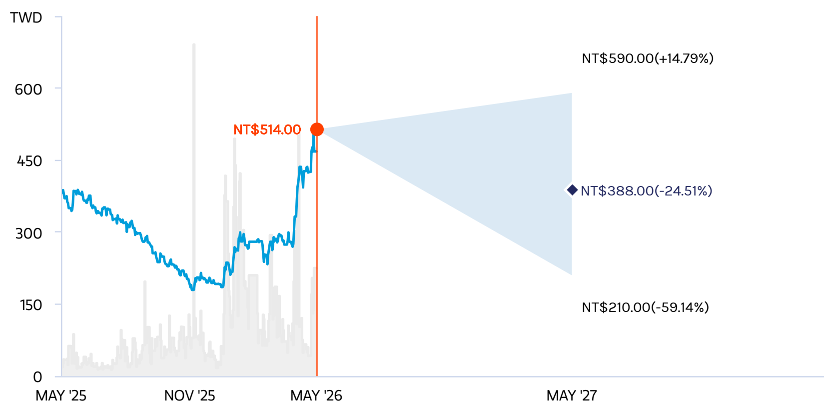


Source: Refinitiv, Morgan Stanley Research

UNDERWEIGHT THESIS

- We have downgraded Silergy in view of its margin miss from foundry cost hikes and weak consumer demand.
- Its server and optical PMIC exposure remains relatively small.
- Our price target implies 23x 2027e P/E. We view valuation as more than full, especially after a strong stock rally.

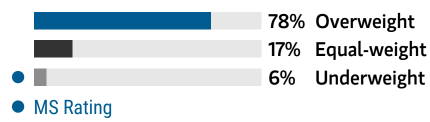
RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power:	*Negative*
Secular Growth:	*Positive*
Technology Diffusion:	*Negative*

View descriptions of Risk Rewards Themes here

BULL CASE NT$590.00

35x 2027e EPS

Substantial market share gains with further margin improvement: Silergy adds high-margin industrial product lines, such as automotive ICs, faster than expected. Meanwhile, legacy product lines show better growth, given faster market share gains and improved local semiconductor demand from China. Revenue CAGR for 2025-28 accelerates to over 25%.

BASE CASE NT$388.00

23x 2027e EPS

Recovering demand in 2026e with margin curbed: Revenue growth momentum may recover in 2026; revenue CAGR for 2025-28e reaches 22%. Pricing environment improves, thanks to easing competition and healthier channel inventory levels. However, owing to manufacturing shift impact, gross margin may decline to <52% in 2026e from 53.8% in 2024.

BEAR CASE NT$210.00

12x 2027e EPS

Competition continues, leading to further price erosion: Revenue growth decelerates because of macro conditions and severe market share loss. Contribution of new products stays sluggish, while gross margin declines to <45% amid weakened product mix and intense price competition in China's driver IC market. Revenue CAGR for 2025-28e falls to <0%.

Risk Reward – Silergy Corp. (6415.TW)

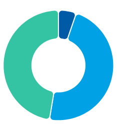
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Revenue from Consumer Segment (NT$, mn)	7,096	8,603	10,486	12,474
Revenue from IT Segment (NT$, mn)	2,150	2,306	2,755	3,278
Revenue from Communication Segment (NT$, mn)	1,505	1,856	2,217	2,637
Revenue from Industrial Segment (NT$, mn)	5,781	7,470	8,924	10,615

INVESTMENT DRIVERS

- LED driver IC market share gain/loss
- New products, e.g., automotive, industrials
- Gross margin improvement/erosion
- China's semiconductor localization demand

GLOBAL REVENUE EXPOSURE



- 0-10% North America
- 40-50% APAC, ex Japan, Mainland China and India
- 40-50% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

4/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Gross margin expansion amid stabilized competition
- A surge in LED demand as pricing drops to the sweet spot
- Greater-than-expected benefits from China's support in local semiconductors
- Higher-than-expected dividend payout ratio

RISKS TO DOWNSIDE

- Lower gross margin amid intensified competition
- Weaker-than-expected benefits from China's support in local semiconductors
- Lower-than-expected dividend payout ratio

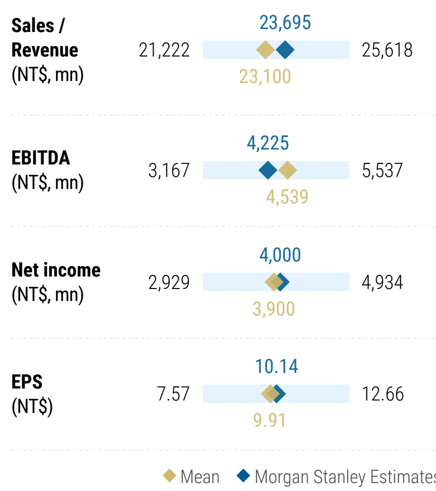
OWNERSHIP POSITIONING

Inst. Owners, % Active 84.7%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

Silergy: Financial Summary

Exhibit 29: Silergy: Financial Summary

Income Statement

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Net sales	18,812	23,695	28,515	33,920
COGS	(9,102)	(11,426)	(13,312)	(15,741)
Gross profit	9,710	12,268	15,203	18,179
Operating expenses	(7,567)	(8,681)	(8,435)	(9,338)
Operating income	**2,144**	**3,588**	**6,767**	**8,840**
Non-operating income	533	634	316	316
Pre-tax income	2,677	4,221	7,083	9,156
Income tax	605	482	708	916
Reported net Income	**2,478**	**4,000**	**6,495**	**8,360**
Adj.wtd.avg.shrs(m)	384	387	387	387
Reported EPS (NT$)	**6.36**	**10.14**	**16.47**	**21.20**
Modelware EPS (NT$)	**6.36**	**10.14**	**16.47**	**21.20**

Balance Sheet

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Cash	9,756	12,146	15,795	20,055
Mkt Securities	7,298	7,298	7,298	7,298
AR/NR	2,354	2,613	3,145	3,741
Inventory	4,434	4,474	5,212	6,163
Other	1,353	1,353	1,353	1,353
Current Assets	**25,196**	**27,884**	**32,804**	**38,610**
Long-term investments	4,782	4,782	4,782	4,782
Fixed assets	5,261	5,331	5,401	5,471
Deffered assets	154	154	154	154
Other assets	8,778	8,778	8,778	8,778
Total Assets	**44,171**	**46,929**	**51,919**	**57,795**
S/T borrowings	662	662	662	662
AP/NP	945	709	826	976
Other ST liabilities	1,967	1,967	1,967	1,967
LT debt	2,844	2,844	2,844	2,844
Other LT liabilities	0	0	0	0
Common shares	971	971	971	971
Total Liabilities	**6,418**	**6,182**	**6,299**	**6,449**
Additional capital	13,882	13,882	13,882	13,882
Retained earning	21,902	24,896	29,769	35,494
Other shareholders' equity	998	998	998	998
Total Equity	**37,753**	**40,748**	**45,620**	**51,345**
Total Liab. & Shrhldr's Equity	**44,171**	**46,929**	**51,919**	**57,795**

E = Morgan Stanley Research Estimates

Source: Morgan Stanley Research, Company Data

Cash Flow Statement

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Cashflow from Operations	**2,304**	**4,102**	**6,043**	**7,735**
Net profits	2,478	4,000	6,495	8,360
Depreciation	580	638	701	771
Working Capital Change	2,651	(535)	(1,153)	(1,396)
Other adjustments	(3,404)	0	0	0
Cashflow from Investing	**(8,534)**	**(708)**	**(771)**	**(841)**
Capex	(2,060)	(708)	(771)	(841)
Change of LT Investment	(4,216)	0	0	0
Change of ST Investment	(887)	0	0	0
Other adjustments	(1,371)	0	0	0
Cashflow from financing	**765**	**(1,005)**	**(1,623)**	**(2,635)**
Increase in L/T debt	1,466	0	0	0
Increase in S/T debt	(84)	0	0	0
Cash Dividend Paid	(927)	(1,005)	(1,623)	(2,635)
Dir& Emp Bonus Paid	0	0	0	0
Issuance of stock	0	0	0	0
Other adjustments	311	0	0	0
Exchange rate adjustment	(982)	0	0	0
Net change in cash	**-6,446**	**2,390**	**3,649**	**4,259**

Financial Ratios

	2025	2026E	2027E	2028E
Growth(%)				
Turnover	1.9	26.0	20.3	19.0
Operating profits	-4.9	67.4	88.6	30.6
Pretax profits	3.6	57.7	67.8	29.3
Net profits	8.4	61.4	62.4	28.7
EPS	7.7	59.4	62.4	28.7
Margins (%)				
Gross Margin	51.6	51.8	53.3	53.6
Operating Margin	11.4	15.1	23.7	26.1
Pretax Margin	14.2	17.8	24.8	27.0
Net Profit	13.2	16.9	22.8	24.6
Return (%)				
ROAE	6.7	10.2	15.0	17.2
ROAA	5.9	8.8	13.1	15.2
Gearing (%)				
Net Debt/Equity	(24.1)	(28.2)	(33.2)	(37.8)
Liabilities/Equity	17.0	15.2	13.8	12.6
Ratios (X)				
Current ratio	7.0	8.4	9.5	10.7
Quick ratio	3.4	4.4	5.5	6.6
Others				
AR/NR Turnover (days)	40	40	40	40
Inventory Turnover (days)	143	143	143	143
AP Turnover (days)	23	23	23	23
Cash Conversion (days)	161	161	161	161

SMIC: Earnings Estimate Revisions

We raise our 2026, 2027, and 2028 EPS estimates 11%, 14%, and 10%, respectively: We factor in the strong 2Q26 revenue and GM guidance, which we believe is driven by the aggressive capacity expansion of advanced node process and improved blended ASP thanks to better product mix and price hike for BCD platform. Therefore, we raise our 2026, 2027, and 2028 revenue estimates 9%, 15%, and 15%. We also raised 2026 and 2027 gross margin assumptions 1.4ppts and 0.1ppts. In the long run, we may have concerns about surging depreciation coming with capacity expansion.

Exhibit 30: SMIC: Earnings estimate revisions

	New 2026e	Old 2026e	Diff.%	New 2027e	Old 2027e	Diff.%	New 2028e	Old 2028e	Diff.%
Net sales	11,938	10,908	9%	14,406	12,567	15%	15,681	13,690	15%
Gross profit	2,646	2,269	17%	3,338	2,905	15%	3,792	3,442	10%
Operating profit	1,727	1,480	17%	2,456	2,029	21%	2,825	2,482	14%
Pretax income	1,975	1,800	10%	2,856	2,469	16%	3,345	3,002	11%
Net income to shareholders	1,659	1,492	11%	2,410	2,121	14%	2,890	2,622	10%
EPS Basic (US$ cent)	20.78	18.68	11%	30.18	26.56	14%	36.19	32.83	10%
EPS for Consensus (US$ cent)	20.74	18.65	11%	30.12	26.51	14%	36.12	32.77	10%
Margins									
Gross margin	22.2%	20.8%		23.2%	23.1%		24.2%	25.1%	
Operating margin	14.5%	13.6%		17.0%	16.1%		18.0%	18.1%	
Pretax margin	16.5%	16.5%		19.8%	19.7%		21.3%	21.9%	
Net margin	13.9%	13.7%		16.7%	16.9%		18.4%	19.2%	

Source: Morgan Stanley Research (e) estimates

Exhibit 31: SMIC: Quarterly financials

(US$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e	2025	2026e	2027e	2028e
Total Revenues	**2,247**	**2,209**	**2,382**	**2,489**	**2,505**	**2,882**	**3,202**	**3,348**	**9,327**	**11,938**	**14,406**	**15,681**
Q/Q %	*1.8%*	*-1.7%*	*7.8%*	*4.5%*	*0.7%*	*15.0%*	*11.1%*	*4.6%*				
Y/Y %	*28.4%*	*16.2%*	*9.7%*	*12.8%*	*11.5%*	*30.4%*	*34.4%*	*34.5%*	*16.2%*	*28.0%*	*20.7%*	*8.9%*
Cost of Sales	**(1,741)**	**(1,759)**	**(1,859)**	**(2,011)**	**(2,002)**	**(2,291)**	**(2,443)**	**(2,556)**	**(7,370)**	**(9,292)**	**(11,067)**	**(11,888)**
Gross Profit	**506**	**450**	**523**	**478**	**504**	**591**	**759**	**793**	**1,957**	**2,646**	**3,338**	**3,792**
Gross Margin	*22.5%*	*20.4%*	*22.0%*	*19.2%*	*20.1%*	*20.5%*	*23.7%*	*23.7%*	*21.0%*	*22.2%*	*23.2%*	*24.2%*
Total Opex	**(196)**	**(299)**	**(172)**	**(180)**	**(256)**	**(233)**	**(250)**	**(180)**	**(847)**	**(919)**	**(883)**	**(968)**
Percent of Revenues	*8.7%*	*13.5%*	*7.2%*	*7.2%*	*10.2%*	*8.1%*	*7.8%*	*5.4%*	*9.1%*	*7.7%*	*6.1%*	*6.2%*
R&D	(35)	(97)	(119)	(26)	(128)	(97)	(107)	(52)	(774)	(843)	(775)	(835)
Percent of Revenues	*1.6%*	*4.4%*	*5.0%*	*1.0%*	*5.1%*	*3.4%*	*3.3%*	*1.6%*	*8.3%*	*7.1%*	*5.4%*	*5.3%*
Sales and Marketing	(11)	(13)	(11)	(8)	(10)	(13)	(14)	(15)	(43)	(51)	(58)	(63)
Percent of Revenues	*0.5%*	*0.6%*	*0.4%*	*0.4%*	*0.4%*	*0.4%*	*0.4%*	*0.4%*	*0.5%*	*0.4%*	*0.4%*	*0.4%*
General and Admin	(149)	(189)	(42)	(146)	(118)	(123)	(128)	(133)	(527)	(503)	(590)	(610)
Percent of Revenues	*6.6%*	*8.6%*	*1.8%*	*5.9%*	*4.7%*	*4.3%*	*4.0%*	*4.0%*	*5.6%*	*4.2%*	*4.1%*	*3.9%*
Operating Income	**310**	**151**	**351**	**299**	**248**	**358**	**509**	**612**	**1,110**	**1,727**	**2,456**	**2,825**
Percent of Revenues	*13.8%*	*6.8%*	*14.7%*	*12.0%*	*9.9%*	*12.4%*	*15.9%*	*18.3%*	*11.9%*	*14.5%*	*17.0%*	*18.0%*
Total Non-operating Income(Loss)	**40**	**10**	**(14)**	**(72)**	**8**	**80**	**80**	**80**	**(37)**	**248**	**400**	**520**
Profit Before Taxes	**349**	**160**	**337**	**227**	**255**	**438**	**589**	**692**	**1,073**	**1,975**	**2,856**	**3,345**
Percent of Revenues	*15.5%*	*7.3%*	*14.2%*	*9.1%*	*10.2%*	*15.2%*	*18.4%*	*20.7%*	*11.5%*	*16.5%*	*19.8%*	*21.3%*
Taxes	(26)	(14)	(22)	(23)	(24)	(35)	(47)	(55)	(84)	(162)	(286)	(334)
Tax Rate	*7.4%*	*8.6%*	*6.4%*	*10.2%*	*9.6%*	*8.0%*	*8.0%*	*8.0%*	*7.9%*	*8.2%*	*10.0%*	*10.0%*
Minority Interests	(135)	(14)	(124)	(31)	(33)	(40)	(40)	(40)	(304)	(153)	(160)	(120)
Loss from equity investment	0	0	0	0	0	0	0	0	0	0	0	0
Net Income to Ordinary Shareholders	**188**	**132**	**192**	**173**	**197**	**363**	**502**	**597**	**685**	**1,659**	**2,410**	**2,890**
Percent of Revenues	*8.4%*	*6.0%*	*8.1%*	*6.9%*	*7.9%*	*12.6%*	*15.7%*	*17.8%*	*7.3%*	*13.9%*	*16.7%*	*18.4%*
Change vs Year Ago	*162%*	*-19%*	*29%*	*61%*	*5%*	*174%*	*162%*	*245%*	*39.0%*	*142.2%*	*45.2%*	*19.9%*
EPS (US$ cts) - Basic	2.35	1.66	2.40	2.16	2.47	4.54	6.29	7.48	8.58	20.78	30.18	36.19
EPS (US$ cts) - Diluted	2.35	1.66	2.40	2.16	2.47	4.53	6.27	7.46	8.56	20.74	30.12	36.12
EPS (HK$ cts) - Diluted	18.44	12.99	18.80	16.95	19.36	35.58	49.23	58.54	67.18	162.71	236.34	283.39

Source: Company data, Morgan Stanley Research (e) estimates

We raise our price target to HK$85 from HK$70: This reflects the changes to our 2026-

28 EPS estimates. We continue to derive our price target through a residual income framework, which we believe best captures the value of the company.

We maintain our cost of equity assumption of 8.6% (beta 1.2, risk-free rate 2% and risk premium 5.5%) as well as our terminal growth rate of 5.0%. Our intermediate growth rate is 16% (unchanged) – we believe the strong capacity expansion in advanced node process can support top-line growth for SMIC, and the company could remain the leader among domestic advanced node semi manufacturing.

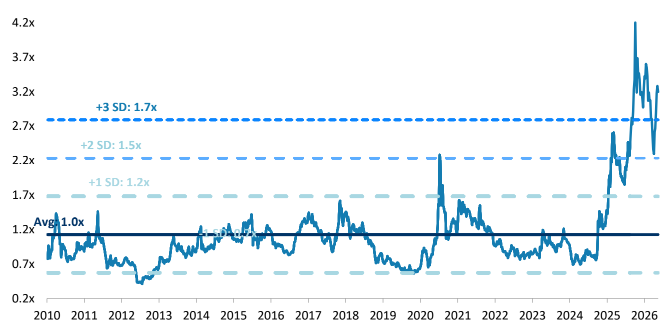
Our bull and bear case scenario values are also increased 21% each, to HK$138 and HK$58, respectively, implying 6.1x and 2.6x 2026e P/B.

Exhibit 32: SMIC: Residual income model

(US$ mn)	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e	2038e
Total Equity	**23,099**	**25,509**	**28,399**	**28,902**	**29,486**	**30,162**	**30,873**	**31,619**	**32,402**	**33,225**	**34,088**	**34,995**	**35,947**
Net Profit	**1,659**	**2,410**	**2,890**	**3,353**	**3,889**	**4,511**	**4,737**	**4,974**	**5,222**	**5,483**	**5,758**	**6,045**	**6,348**
ROAE	**7.5%**	**9.9%**	**10.7%**	**11.7%**	**13.3%**	**15.1%**	**15.5%**	**15.9%**	**16.3%**	**16.7%**	**17.1%**	**17.5%**	**17.9%**
Residual Income	(246)	304	541	881	1,365	1,924	2,088	2,259	2,439	2,628	2,826	3,034	3,253
Spread	**-1.1%**	**1.3%**	**2.1%**	**3.1%**	**4.7%**	**6.5%**	**6.9%**	**7.3%**	**7.7%**	**8.1%**	**8.5%**	**8.9%**	**9.3%**
Ending Equity Capital	23,099												
PV of Forecast Period	11,490												
PV of Continuing Value	52,182												
Equity Value	**86,771**												
No. of Shares	8,002												
FX	0.127												
Price Target (HK$)	**85**												

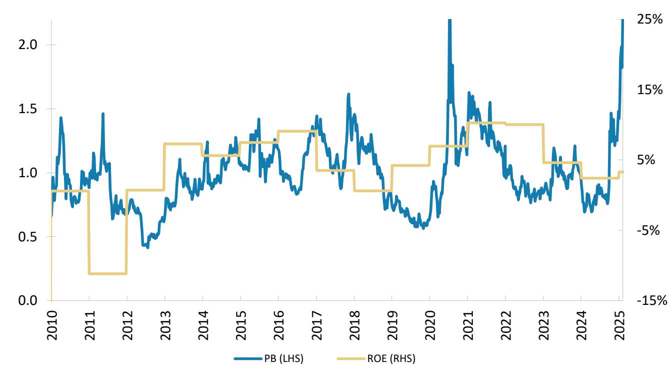
Source: Morgan Stanley Research (e) estimates

Exhibit 33: SMIC: One-year forward P/B



Source: Company data, FactSet, Morgan Stanley Research estimates

Exhibit 34: SMIC: One-year forward P/B vs. ROE trend



Source: Company data, FactSet, Morgan Stanley Research estimates

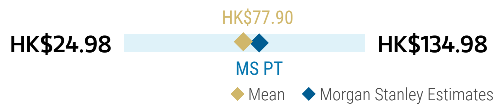
Risk Reward – SMIC (0981.HK)

AI spillover story

PRICE TARGET HK$85.00

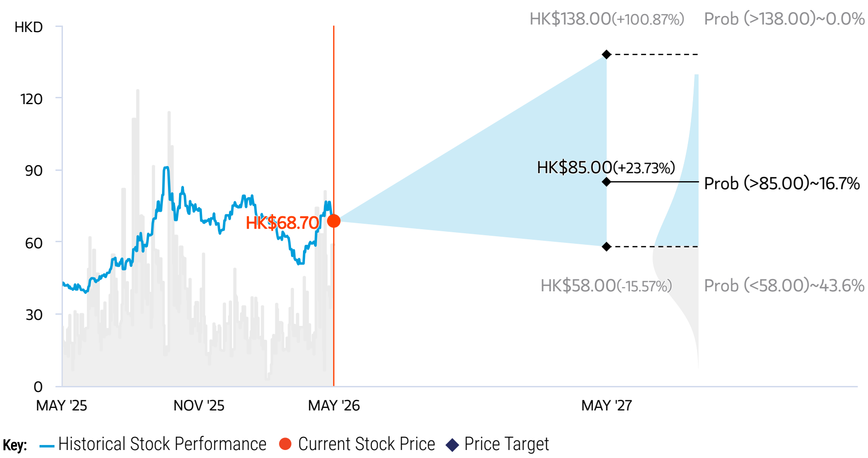
Base-case, residual income model. Key assumptions: cost of equity 8.6% (beta 1.2, risk-free rate 2% and risk premium 5.5%), intermediate growth rate 16%, terminal growth rate 5%, and cash dividend payout ratio 85%.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)

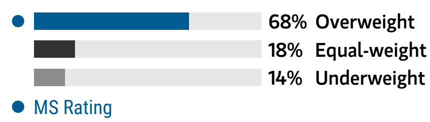


Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 18 May 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology here

OVERWEIGHT THESIS

- We believe the AI cycle is increasingly spilling over from leading-edge computing into mature node companion silicon (AI-driven PMIC demand supporting specialty node utilization).
- The company has framed its AI-related opportunity around power management, edge AI logic, connectivity, etc., rather than the AI accelerator itself.
- This distinction matters: the incremental demand described by management is not a direct GPU story, but a mature node and specialty process story.
- Our price target implies 3.8x 2026e P/B and 52x 2026e P/E, which we believe still appears attractive given SMIC's uniqueness in domestic AI chip production.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era:	*Positive*
Special Situation:	*Positive*
Technology Diffusion:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE HK$138.00

6.1x 2026e BVPS

Better profits and growth than historical trends: Domestic AI demand remains strong and push SMIC's utilization rate to remain high for advanced node process. 40nm and 28nm demand grows more strongly than expected, while SMIC gains share in the Chinese market, thanks to smooth capacity expansion and faster 28nm node development. Higher blended ASP and UTR drives 2026 gross margin forecast to exceed 30%.

BASE CASE HK$85.00

3.8x 2026e BVPS

Beneficiary of strong AI demand in China: SMIC maintains its strategic position in China, especially with its advanced node offerings, along with the largest capacity in mature nodes domestically. Maintaining aggressive capex may lead to high depreciation, leading to a 2026 gross margin forecast of 22%.

BEAR CASE HK$58.00

2.6x 2026e BVPS

Domestic AI semi and traditional semi demand slows down: If local CSPs buy more AI chips from US vendors like NVIDIA and AMD and local AI processors lose market share, SMIC's utilization rate may drop for advanced node process. The global economy deteriorates significantly, hurting global semi demand. China's consumption stimulus stops, resulting in demand that is not strong enough to boost 40nm demand, intensifying competition. 28nm overcapacity issue persists.

Risk Reward – SMIC (0981.HK)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Wafer shipment (k 8 inch equivalent)	9,708.4	10,889.7	2,672.8	2,918.8
Wafer ASP (USD) (US$)	914	1,052	1,033	1,182
Utilization rate (%)	0.8	0.8	0.8	0.8

CATALYST CALENDAR

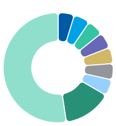
Date	Event
26 Jun 2026 - 30 Jun 2026	Semiconductor Manufacturing International Corp Annual Shareholders Meeting

Source: Refinitiv, Morgan Stanley

INVESTMENT DRIVERS

- Global semi growth
- Market share in mature nodes, with semi demand from China
- IDM outsourcing

GLOBAL REVENUE EXPOSURE



- 0-10% Europe ex UK
- 0-10% India
- 0-10% Japan
- 0-10% Latin America
- 0-10% MEA
- 0-10% North America
- 0-10% UK
- 10-20% APAC, ex Japan, Mainland China and India
- 50-60% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

5/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Global and Chinese semi demand strengthens.
- Pricing competition eases, helping margins.
- Technological breakthroughs come faster than expected, shortening transition period.
- Utilization rate, product mix and ASP are better than expected.

RISKS TO DOWNSIDE

- Global and Chinese semi demand weakens.
- Pricing competition intensifies, eroding margins.
- Utilization rate, product mix and ASP trend are worse than expected.

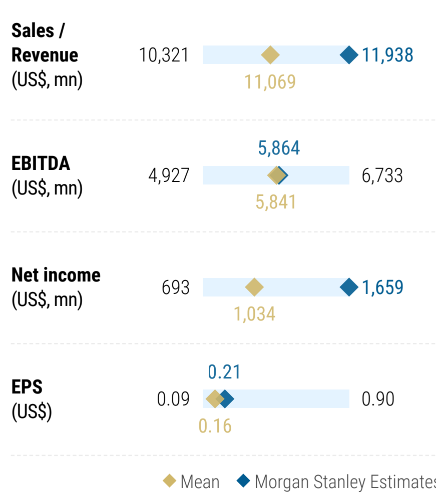
OWNERSHIP POSITIONING

Inst. Owners, % Active 74.3%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

SMIC: Financial Summary

Exhibit 35: SMIC: Financial summary

Income Statement, Year end Dec.

(US$ mn)	2025	2026e	2027e	2028e
Turnover	**9,327**	**11,938**	**14,406**	**15,681**
YoY Growth	*16%*	*28%*	*21%*	*9%*
Less: COGS	(7,370)	(9,292)	(11,067)	(11,888)
Variable costs	(4088)	(5337)	(6301)	(6743)
Depreciation & amort	(3282)	(3955)	(4766)	(5146)
Gross profit	**1957**	**2646**	**3338**	**3792**
% margin	*21%*	*22%*	*23%*	*24%*
Operating Expenses:	(847)	(919)	(883)	(968)
R&D	(774)	(843)	(775)	(835)
Sales and Marketing	(43)	(51)	(58)	(63)
General and Admin	(527)	(503)	(590)	(610)
Stock based compen	0	0	0	0
Other non-recurring charges	497	479	540	540
Operating Profit	**1110**	**1727**	**2456**	**2825**
% margin	*12%*	*14%*	*17%*	*18%*
Forex gain	29	21	0	0
Interest expense	(373)	(362)	(360)	(360)
Interest income	398	428	480	480
Other income (expense)	(80)	6	0	0
Pretax Profit	**1073**	**1975**	**2856**	**3345**
% margin	*12%*	*17%*	*20%*	*21%*
Tax	(84)	(162)	(286)	(334)
Net Income b/f div	**685**	**1659**	**2410**	**2890**
Deemed Dividend	0	0	0	0
Net Income a/f div	**685**	**1659**	**2410**	**2890**
EPS (US$ cents)	**8.56**	**20.74**	**30.12**	**36.12**
EPS (HK$ cents)	**67.18**	**162.71**	**236.34**	**283.39**
Key Operating Data				
Capacity Utilisation(%)	80%	77%	79%	82%
Blended ASP (US$)	914	955	1033	1073

Balance sheet

(US$ mn)	2025	2026e	2027e	2028e
Assets				
Cash & Cash Equivalent	**5,873**	**3,914**	**1,865**	**4,226**
Receivables	1,433	1,249	1,507	1,641
Inventories	3,630	4,176	4,974	5,343
Prepaid expenses	0	0	0	0
Other current assets	4,690	6,870	8,290	9,024
Total current assets	**15,625**	**16,209**	**16,636**	**20,234**
Acquired Intangible Assets	20	20	20	20
Land Use rights	0	0	0	0
Fixed assets	32,558	38,182	42,364	43,104
Total non-current assets	**36,646**	**42,270**	**46,453**	**47,192**
Total Assets	**52,271**	**58,480**	**63,089**	**67,426**
Liabilities and Shareholders' Equity				
Trades Payable	3,592	7,322	8,721	9,368
Current portion of capital lease	0	0	0	0
STB + Current portion of LTD	2,593	2,593	2,593	2,593
Accrued op. expenses	410	410	410	410
Other current liabilities	189	189	189	189
Total current liabilities	**6,817**	**10,547**	**11,946**	**12,593**
Long-term debts	9,995	10,495	10,995	11,495
Bond Payable	0	0	0	0
Other Loans	378	397	397	397
Other liabilities	61	61	61	61
Total non-current liabilities	10,434	10,953	11,453	11,953
Total Liabilities	**17,251**	**21,500**	**23,399**	**24,546**
Share capital	32	32	32	32
Subscription receivable	0	0	0	0
Paid in capital	14,395	14,395	14,395	14,395
Deferrred Stock Compensation	0	0	0	0
Accumulated other income	155	155	155	155
Retained income (deficit)	6,858	8,518	10,928	13,818
Total Equity	**21,440**	**23,099**	**25,509**	**28,399**
Total Liab. & Equity	**52,271**	**58,480**	**63,089**	**67,426**

Cash Flow Statement

(US$ mn)	2025	2026e	2027e	2028e
Pretax profit adjusted	1,073	1,975	2,856	3,345
Depreciation and amort	3,810	4,629	5,416	6,079
Tax	(84)	(162)	(286)	(334)
Others	2,197	0	0	0
Fund From Operations	**6,692**	**6,289**	**7,826**	**8,969**
(Inc)/dec in working capital	**(2,942)**	**1,187**	**(1,078)**	**(589)**
Operating Cash flow	**3,750**	**7,476**	**6,749**	**8,380**
Investing Activities	0	0	0	0
Net Fixed assets Additions	(8,098)	(10,253)	(9,598)	(6,819)
Investments	0	0	0	0
Net CF from investments	**(8,098)**	**(10,253)**	**(9,598)**	**(6,819)**
Change in capital	129	0	0	0
Dividend	0	0	0	0
Minorities & others	2,325	300	300	300
Increase/ (repayment) of debt	2,053	500	500	500
Net CF from finances	**4,506**	**800**	**800**	**800**
Change in Cash	**158**	**(1,977)**	**(2,050)**	**2,361**
Beginning Cash	6,312	5,891	3,914	1,865
Ending Cash	**5,891**	**3,914**	**1,865**	**4,226**

Key Ratios

(US$ mn)	2025	2026e	2027e	2028e
Return (%)				
ROAA	1%	3%	4%	4%
ROAE	3%	7%	9%	10%
OP. ATO	0.2x	0.3x	0.3x	0.3x
Gearing (x)				
Net debt/ Equity	0.2x	0.3x	0.4x	0.3x
Current Ratio	2.3x	1.5x	1.4x	1.6x
Quick Ratio	1.1x	0.5x	0.3x	0.5x

Source: Company Data, Morgan Stanley Research;
e = Morgan Stanley Research estimates

Hua Hong: Estimate Revisions

We lower our 2026 EPS estimate 7% but raise our estimates 15% for 2027 and 20% for 2028: We factor in the strong demand for AI-related power management semis and potential price hike for the BCD platform, where Hua Hong is highly capable with its specialty technology. Although consumer-related products may face headwinds given memory price hikes, we believe demand from AI companion products should mostly offset the revenue decline. We also see promising capacity and shipment growth trends, along with Fab 9 ramp-up. Minority interests could still remain high, which would improve earnings.

Exhibit 36: Hua Hong: Earnings estimate revision summary

(US$ mn)	New 2026e	Old 2026e	Diff.%	New 2027e	Old 2027e	Diff.%	New 2028e	Old 2028e	Diff.%
Net sales	3,053	3,117	-2%	3,593	3,460	4%	3,937	3,700	6%
Gross profit	540	560	-4%	800	684	17%	978	809	21%
Operating profit	242	280	-13%	487	372	31%	648	481	35%
Pretax income	185	240	-23%	447	332	35%	608	441	38%
Net income to shareholders	375	404	-7%	580	502	15%	737	615	20%
Basic EPS (US$ cents)	**21.83**	**23.51**	**-7%**	**33.74**	**29.23**	**15%**	**42.90**	**35.77**	**20%**
EPS for consensus (US$ cents)	**21.76**	**23.43**	**-7%**	**33.63**	**29.13**	**15%**	**42.75**	**35.65**	**20%**
Margins									
Gross margin	17.7%	18.0%		22.3%	19.8%		24.8%	21.9%	
Operating margin	7.9%	9.0%		13.5%	10.8%		16.5%	13.0%	
Pretax margin	6.1%	7.7%		12.4%	9.6%		15.5%	11.9%	
Net margin	12.3%	13.0%		16.1%	14.5%		18.7%	16.6%	

Source: Morgan Stanley Research (e) estimates

We raise our price target to HK$118 from HK$88: This reflects the changes to our 2026-28 EPS estimates. We also raise the dividend payout ratio to 95% from 91% to reflect potential dividends after heavy investment in recent years.

We continue to derive our price target from the base case scenario value in our residual income model, which we believe best captures the value of the company. All our other key assumptions are unchanged, including a 9.2% cost of equity (beta 1.20, risk-free rate 2% and risk premium 6%), a 16.5% intermediate growth rate, and a terminal growth rate of 5.0%. Our base case price target implies 3.3x our 2026e BVPS.

Our bull and bear case scenario values increase from HK$119 and HK$47 to HK$160 and HK$63, respectively, implying 4.4x and 1.7x our 2026e BVPS.

Exhibit 37: Hua Hong: Residual income model

(US$ mn)	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e	2038e
Total Equity	**7,974**	**9,460**	**10,052**	**10,177**	**10,322**	**10,491**	**10,668**	**10,854**	**11,050**	**11,256**	**11,471**	**11,698**	**11,935**
Net Profit	**375**	**580**	**737**	**859**	**1,000**	**1,165**	**1,224**	**1,285**	**1,349**	**1,417**	**1,487**	**1,562**	**1,640**
ROAE	**5.1%**	**6.7%**	**7.6%**	**8.5%**	**9.8%**	**11.2%**	**11.6%**	**11.9%**	**12.3%**	**12.7%**	**13.1%**	**13.5%**	**13.9%**
Residual Income	(268)	(203)	(156)	(71)	57	206	248	399	447	718	775	950	1,132
Spread	**-4.1%**	**-2.5%**	**-1.6%**	**-0.7%**	**0.6%**	**2.0%**	**2.4%**	**2.7%**	**3.1%**	**3.5%**	**3.9%**	**4.3%**	**4.7%**
Ending Equity Capital	7,974												
PV of Forecast Period	1,392												
PV of Continuing Value	16,698												
Equity Value	**26,065**												
No. of Shares	1,718												
FX	7.8												
Price Target (HK$)	**118**												

Source: Morgan Stanley Research (e) estimates

Exhibit 38: Hua Hong: P/B

Source: Morgan Stanley Research (e) estimates

Exhibit 39: Hua Hong: Quarterly financials

(US$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e	2025	2026e	2027e	2028e
Total Revenues	**541**	**566**	**635**	**660**	**661**	**694**	**822**	**877**	**2,402**	**3,053**	**3,593**	**3,937**
Q/Q %	*0.3%*	*4.6%*	*12.2%*	*3.9%*	*0.2%*	*4.9%*	*18.5%*	*6.7%*				
Y/Y %	*17.6%*	*18.3%*	*20.7%*	*22.4%*	*22.2%*	*22.5%*	*29.4%*	*32.9%*	*19.9%*	*27.1%*	*17.7%*	*9.6%*
Cost of Sales	**(491)**	**(504)**	**(549)**	**(574)**	**(575)**	**(587)**	**(663)**	**(688)**	**(2,119)**	**(2,513)**	**(2,793)**	**(2,959)**
Gross Profit	**50**	**62**	**86**	**85**	**86**	**106**	**158**	**189**	**283**	**540**	**800**	**978**
Gross Margin	*9.2%*	*10.9%*	*13.5%*	*13.0%*	*13.0%*	*15.3%*	*19.3%*	*21.6%*	*11.8%*	*17.7%*	*22.3%*	*24.8%*
Total Opex	**(82)**	**(70)**	**(66)**	**(89)**	**(81)**	**(68)**	**(64)**	**(84)**	**(307)**	**(297)**	**(313)**	**(330)**
Percent of Revenues	*15.2%*	*12.3%*	*10.4%*	*13.5%*	*12.2%*	*9.9%*	*7.8%*	*9.6%*	*12.8%*	*9.7%*	*8.7%*	*8.4%*
R&D	(95)	(95)	(98)	(126)	(103)	(105)	(100)	(120)	(415)	(428)	(470)	(510)
Percent of Revenues	*17.5%*	*16.8%*	*15.4%*	*19.2%*	*15.6%*	*15.1%*	*12.2%*	*13.7%*	*17.3%*	*14.0%*	*13.1%*	*13.0%*
Sales and Marketing	(2)	(3)	(2)	(4)	(2)	(3)	(4)	(4)	(11)	(14)	(18)	(20)
Percent of Revenues	*0.5%*	*0.5%*	*0.5%*	*0.5%*	*0.5%*	*0.5%*	*0.5%*	*0.5%*	*0.5%*	*0.5%*	*0.5%*	*0.5%*
General and Admin	15	28	34	41	25	40	40	40	119	145	175	200
Percent of Revenues	*2.8%*	*5.0%*	*5.4%*	*6.3%*	*3.8%*	*5.8%*	*4.9%*	*4.6%*	*4.9%*	*4.8%*	*4.9%*	*5.1%*
Operating Income	**(32)**	**(8)**	**20**	**(3)**	**6**	**38**	**94**	**105**	**(24)**	**242**	**487**	**648**
Percent of Revenues	*-5.9%*	*-1.4%*	*3.1%*	*-0.5%*	*0.8%*	*5.4%*	*11.5%*	*11.9%*	*-1.0%*	*7.9%*	*13.5%*	*16.5%*
Total Non-operating Income(Loss)	**(23)**	**(18)**	**(16)**	**(7)**	**(28)**	**(10)**	**(10)**	**(10)**	**(65)**	**(57)**	**(40)**	**(40)**
Profit Before Taxes	**(55)**	**(26)**	**3**	**(11)**	**(22)**	**28**	**85**	**95**	**(0)**	**(3)**	**1**	**0**
Percent of Revenues	*22.8%*	*-23.0%*	*-90.8%*	*-88.2%*	*-60.3%*	*-208.1%*	*2564.0%*	*-996.0%*	*0.0%*	*0.0%*	*0.0%*	*0.0%*
Taxes	3	(7)	(10)	(8)	5	3	8	(14)	(22)	2	(67)	(91)
Tax Rate	*5.9%*	*-27.6%*	*326.6%*	*-76.2%*	*21.5%*	*-10.0%*	*-10.0%*	*15.0%*	*-25.2%*	*-0.9%*	*15.0%*	*15.0%*
Minority Interests	56	41	33	36	38	50	50	50	166	188	200	220
Net Income	**4**	**8**	**26**	**17**	**21**	**81**	**143**	**131**	**55**	**375**	**580**	**737**
Percent of Revenues	*0.7%*	*1.4%*	*4.1%*	*2.6%*	*3.2%*	*11.6%*	*17.4%*	*14.9%*	*2.3%*	*12.3%*	*16.1%*	*18.7%*
Change vs Year Ago	*-88%*	*19%*	*-43%*	*-169%*	*458%*	*913%*	*456%*	*649%*	*-5.6%*	*583.5%*	*54.5%*	*27.1%*
EPS (US$ cts) - Basic	0.22	0.46	1.50	1.02	1.22	4.69	8.32	7.60	3.19	21.83	33.74	42.90
EPS (US$ cts) - Diluted	0.22	0.46	1.49	1.01	1.21	4.67	8.29	7.58	3.18	21.76	33.63	42.75
EPS (HK$ cts) - Diluted	1.69	3.59	11.62	7.88	9.45	36.39	64.59	59.03	24.79	169.47	261.91	332.98

Source: Company data, Morgan Stanley Research (e) estimates

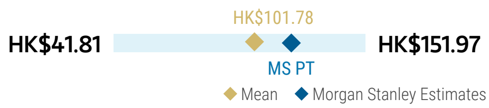
Risk Reward – Hua Hong Semiconductor Ltd (1347.HK)

Valuation looks fair after the rally

PRICE TARGET HK$118.00

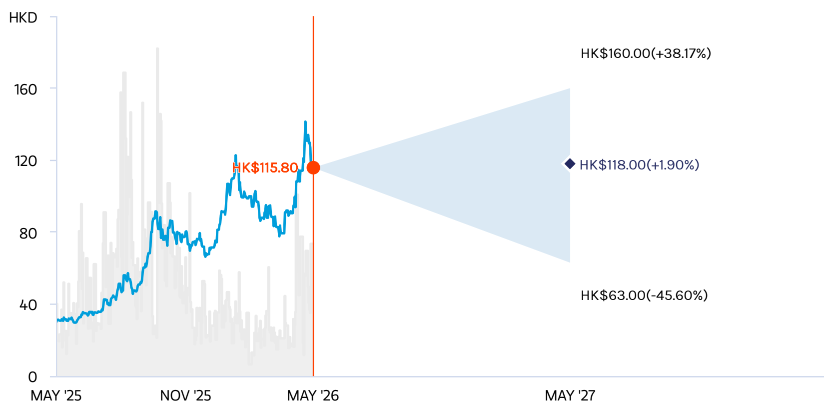
Base case, residual income (RI) analysis. Our RI model is based on the following assumptions: cost of equity of 9.2% (2.0% risk-free rate, 6% risk premium and 1.20 beta), 16.5% medium-term growth rate, 5.0% terminal growth rate, and dividend payout ratio of 95%.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



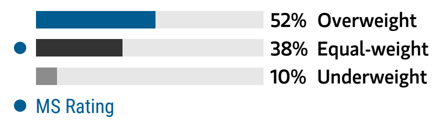
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- Hua Hong's AI read-through is concentrated in PMIC, largely built on its BCD platform and specialty technology.
- Demand is driven by AI servers, automotive electronics, and robotics, while consumer remains soft. PMIC already grew in 1Q but was capacity-constrained.
- NOR Flash is benefiting from memory-cycle spillover.
- Power discretes are healthy, but less pricing-sensitive.
- Valuation is fair relative to our increased price target.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Disruption:	*Positive*
Pricing Power:	*Negative*

View descriptions of Risk Rewards Themes here

BULL CASE HK$160.00

4.4x 2026e BVPS

1) We expect Hua Hong's 8-inch business to remain fully utilized (utilization rate >100%). 2) Domestic customers gain more share than expected from overseas vendors and outsource more orders to Hua Hong. 3) Hua Hong's 12-inch capacity business ramps up sooner than expected, expanding gross margin rapidly. 4) ASP trend starts showing strong rebound.

BASE CASE HK$118.00

3.3x 2026e BVPS

1) Domestic customers gain share from overseas vendors and outsource orders to Hua Hong. 2) Hua Hong's power semi and MCU customers are coming back, while CIS and NOR are improving. It continues to diversify its product portfolio. 3) Gross margin should face pressure along with the capacity ramp-up of Fab 9.

BEAR CASE HK$63.00

1.7x 2026e BVPS

1) We expect Hua Hong's 8-inch business utilization rate to drop to 80%. 2) Domestic customers lose more share than expected from overseas vendors and do not outsource orders to Hua Hong. 3) Hua Hong's 12-inch capacity logic (CIS) business declines more than expected, as capacity cannot be replaced by other applications. 4) ASP trend continues to deteriorate.

Risk Reward – Hua Hong Semiconductor Ltd (1347.HK)

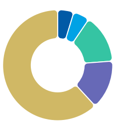
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Wafer shipment (k 8 inch)	5,268	6,395	7,599	8,395
Wafer ASP (USD)	442.9	0.0	474.9	454.9
Utilization rate (%) (%)	96.3	96.1	94.8	93.1

INVESTMENT DRIVERS

- Power semi industry growth, thanks to secular trends such as electric vehicles, 5G and Internet of Things
- Customers' market share and outsourcing ratio
- Fab utilization rate, driven by specialty semi demand

GLOBAL REVENUE EXPOSURE



- 0-10% Europe ex UK
- 0-10% Japan
- 10-20% APAC, ex Japan, Mainland China and India
- 10-20% North America
- 60-70% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

4/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Fab utilization and wafer ASP are healthier than expected.
- Local customers gain share faster, continue to work with Hua Hong.
- Electric vehicle penetration is faster.

RISKS TO DOWNSIDE

- Potential price hike does not materialize.
- Other Chinese players build 8-inch capacity aggressively.
- Applications migrate to 12-inch faster than expected.
- Wuxi fab scales up slowly or struggles to get new customers.

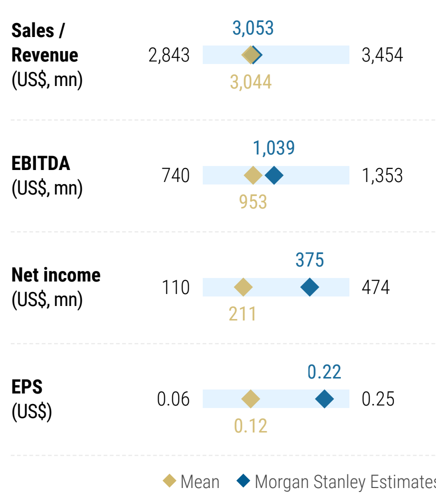
OWNERSHIP POSITIONING

Inst. Owners, % Active 45.1%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

Financial summary

Exhibit 40: Hua Hong: Financial summary [fix 2026e blended ASP cell] done

Income Statement

(US$ mn)	2025	2026e	2027e	2028e
Turnover	**2,402**	**3,053**	**3,593**	**3,937**
YoY Growth	*20%*	*27%*	*18%*	*10%*
Less: COGS	(2,119)	(2,513)	(2,793)	(2,959)
Variable costs	(1443)	(1717)	(1976)	(2126)
Depreciation & amort	(676)	(796)	(817)	(833)
Gross profit	**283**	**540**	**800**	**978**
% margin	*12%*	*18%*	*22%*	*25%*
Operating Expenses:	(307)	(297)	(313)	(330)
R&D and Admin	(415)	(428)	(470)	(510)
Sales and Marketing	(11)	(14)	(18)	(20)
Other income (expenses)	119	145	175	200
Stock based compen	0	0	0	0
Other non-recurring charges	0	0	0	0
Operating Profit	**(24)**	**242**	**487**	**648**
% margin	*-1%*	*8%*	*14%*	*16%*
Investment gain (loss)	0	0	0	0
Interest Income (expense)	(68)	(58)	(40)	(40)
Other income (expense)	0	0	0	0
Associates	4	1	0	0
Pretax Profit	**(89)**	**185**	**447**	**608**
% margin	*-4%*	*6%*	*12%*	*15%*
Tax	(22)	2	(67)	(91)
Net Income (Loss), Cont Ops	**(111)**	**187**	**380**	**517**
Minority Interests	166	188	200	220
Net Income to Ord.	**55**	**375**	**580**	**737**
EPS (US$ cents)	**3.2**	**21.8**	**33.6**	**42.8**
EPS (HK$ cents)	**24.8**	**169.5**	**261.9**	**333.0**
Key Operating Data				
Capacity Utilisation(%)	96%	96%	96%	95%
Total Wafers Shipped (8-inch eq.)	4,629	5,268	6,395	7,599
Blended ASP (US$)	$443	$462	$475	$455

Cash Flow Statement

(US$ mn)	2,025.0	2026e	2027e	2028e
Net Income Current	55	375	580	737
Depreciation	731	866	888	906
Cash Flow from Operations	**785**	**1,241**	**1,468**	**1,643**
Change in Working Capital	(112)	(268)	(218)	(138)
Others	(23)	111	111	111
Operating Cash flow	**650**	**1,083**	**1,361**	**1,616**
(Purchase) of FA	(2,086)	(2,737)	(3,175)	(1,067)
Changes in L-T Inv.	(10)	0	0	0
Changes in S-T Inv.	0	0	0	0
Oth.Adjust-InvAct.	377	0	0	0
Net CF from investments	(1,719)	(2,737)	(3,175)	(1,067)
Change in Debt	993	3,000	1,000	3,000
Issuance of shares	49	1,000	1,000	0
Dividend Paid	(15)	(14)	(94)	(145)
Others	472	(111)	(111)	(111)
Net CF from finances	**1,499**	**3,876**	**1,796**	**2,744**
Change in Cash	**502**	**2,221**	**(19)**	**3,293**
Beginning Cash	4,459	4,961	7,182	7,164
Ending Cash	**4,961**	**7,182**	**7,164**	**10,457**

Balance sheet

(US$ mn)	2025	2026e	2027e	2028e
Assets				
Cash & Cash Equivalent	**4,961**	**7,182**	**7,164**	**10,457**
Receivables	282	358	422	462
Inventories	544	646	718	760
Short term investments	0	0	0	0
Other current assets	819	971	1,098	1,178
Total current assets	**6,607**	**9,158**	**9,401**	**12,857**
Long term investments	0	0	0	0
Intangible assets	36	36	36	36
Fixed assets	6,676	8,548	10,835	10,996
Total non-current assets	1,135	1,135	1,135	1,135
Total Assets	**14,454**	**18,877**	**21,406**	**25,024**
Liabilities and Shareholders' Equity				
Trades Payable	330	392	435	461
Other payables and accrurals	1,009	1,009	1,009	1,009
STB + Current portion of LTD	404	404	404	404
Income taxes payable	15	15	15	15
Other current liabilities	99	99	99	99
Total current liabilities	**1,856**	**1,918**	**1,961**	**1,987**
Capital lease	0	0	0	0
Long-term debts	2,787	5,787	6,787	9,787
Other liabilities	646	646	646	646
Total non-current liabilities	3,433	6,433	7,433	10,433
Total Liabilities	**5,290**	**8,351**	**9,395**	**12,421**
Share capital	4,987	5,987	6,987	6,987
Non-controlling interests	2,552	2,552	2,552	2,552
Total Reserves	1,625	1,987	2,473	3,065
Total Equity	**6,613**	**7,974**	**9,460**	**10,052**
Total Liab. & Equity	**14,454**	**18,877**	**21,406**	**25,024**

Key Ratios

(US$ mn)	2025	2026e	2027e	2028e
Return (%)				
ROI	0.7%	3.1%	4.1%	4.3%
ROAA	0.4%	2.3%	2.9%	3.2%
ROAE	0.9%	5.1%	6.7%	7.6%
Gearing				
Net debt/ Equity	12%	20%	27%	23%
Interest cover	-0.3x	4.2x	12.2x	16.2x
Working Capital (days)				
AR Days	43	43	43	43
Inventory Days	94	94	94	94
Payable Days	57	57	57	57
BVPS (HKD)	**30.0**	**36.1**	**42.9**	**45.6**

Source: Company Data, Morgan Stanley Research;
e = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research (e) estimates

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Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/18/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US$63.67
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb438.75
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT$147.50
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT$4,590.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT$510.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,216.99
Global Unichip Corp (3443.TW)	O (07/27/2024)	NT$4,740.00
GlobalWafers Co Ltd (6488.TWO)	O (09/19/2025)	NT$698.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT$544.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK$115.80

Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK$479.40
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT$295.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb130.80
MediaTek (2454.TW)	O (11/28/2025)	NT$3,400.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb728.47
Nanya Technology Corp. (2408.TW)	E (03/20/2026)	NT$305.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb597.97
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb99.79
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT$2,735.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb105.35
Silergy Corp. (6415.TW)	U (05/18/2026)	NT$514.00
SMIC (0981.HK)	O (10/21/2025)	HK$68.70
TSMC (2330.TW)	O (02/07/2022)	NT$2,240.00
UMC (2303.TW)	O (05/18/2026)	NT$111.00
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT$159.00
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT$450.50
Daisy Dai, CFA		
ASMPT Ltd (0522.HK)	O (07/24/2025)	HK$176.40
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb62.06
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT$145.00
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb94.54
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb31.05
Innoscience (2577.HK)	E (10/13/2025)	HK$71.45
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb58.33
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK$38.74
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb149.84
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb123.60
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb78.33
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb40.09
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb81.40
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT$1,135.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT$1,390.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT$16,150.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT$137.50
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb183.79
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb400.00
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT$160.00
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK$440.80
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb247.99
Novatek (3034.TW)	U (02/04/2026)	NT$475.50
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT$148.50
Parade Technologies Ltd (4966.TWO)	E (01/30/2026)	NT$757.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT$62.00
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT$560.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb64.86
Winbond Electronics Corp (2344.TW)	E (03/20/2026)	NT$130.50
WPG Holdings (3702.TW)	O (03/16/2026)	NT$110.50
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT$263.50
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb157.78
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb592.22
Tiffany Yeh		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT$1,135.00

FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT$804.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US$19.06
Hon Precision (7769.TW)	O (04/17/2026)	NT$6,655.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT$6,135.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US$259.99
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT$10,240.00

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* Historical prices are not split adjusted.